wrangle_act

August 19, 2022

1 Project: Wrangling and Analyze Data

1.1 Table of content

1.2 Data Gathering

In the cell below, gather **all** three pieces of data for this project and load them in the notebook. **Note:** the methods required to gather each data are different. 1. Directly download the WeRate-Dogs Twitter archive data (twitter_archive_enhanced.csv)

In [ ]:

In [131]:
```
#importing libraries for analysis
import pandas as pd
import seaborn as sb
import matplotlib.pyplot as plt
import requests
import json
import numpy as np

%matplotlib inline
```

reading twitter archive

```
In [132]: df_archive = pd.read_csv('twitter-archive-enhanced.csv')   #reading in twitter-archive
```

```
In [133]: df_archive.head()
```

```
Out[133]:             tweet_id  in_reply_to_status_id  in_reply_to_user_id  \
          0  892420643555336193                    NaN                  NaN
          1  892177421306343426                    NaN                  NaN
          2  891815181378084864                    NaN                  NaN
          3  891689557279858688                    NaN                  NaN
          4  891327558926688256                    NaN                  NaN

                            timestamp  \
          0  2017-08-01 16:23:56 +0000
          1  2017-08-01 00:17:27 +0000
          2  2017-07-31 00:18:03 +0000
          3  2017-07-30 15:58:51 +0000
          4  2017-07-29 16:00:24 +0000

                                                        source  \
          0  <a href="http://twitter.com/download/iphone" r...
          1  <a href="http://twitter.com/download/iphone" r...
          2  <a href="http://twitter.com/download/iphone" r...
          3  <a href="http://twitter.com/download/iphone" r...
          4  <a href="http://twitter.com/download/iphone" r...

                                                          text  retweeted_status_id  \
          0  This is Phineas. He's a mystical boy. Only eve...                  NaN
          1  This is Tilly. She's just checking pup on you...                  NaN
          2  This is Archie. He is a rare Norwegian Pouncin...                  NaN
          3  This is Darla. She commenced a snooze mid meal...                  NaN
          4  This is Franklin. He would like you to stop ca...                  NaN

             retweeted_status_user_id retweeted_status_timestamp  \
          0                       NaN                        NaN
          1                       NaN                        NaN
          2                       NaN                        NaN
          3                       NaN                        NaN
          4                       NaN                        NaN

                                                 expanded_urls  rating_numerator  \
          0  https://twitter.com/dog_rates/status/892420643...                13
          1  https://twitter.com/dog_rates/status/892177421...                13
          2  https://twitter.com/dog_rates/status/891815181...                12
          3  https://twitter.com/dog_rates/status/891689557...                13
          4  https://twitter.com/dog_rates/status/891327558...                12

             rating_denominator      name doggo floofer pupper puppo
          0                  10   Phineas  None    None   None  None
```

```
1                   10     Tilly  None    None   None  None
2                   10    Archie  None    None   None  None
3                   10     Darla  None    None   None  None
4                   10  Franklin  None    None   None  None
```

2. Use the Requests library to download the tweet image prediction (image_predictions.tsv)

```
In [134]: # querring the url
          url = 'https://d17h27t6h515a5.cloudfront.net/topher/2017/August/599fd2ad_image-predict

          response = requests.get(url)
```

```
In [135]: response #checking if response is good
```

```
Out[135]: <Response [200]>
```

```
In [136]: #creating image_prediction as tsv file and reading response with a context manager as
          with open('image_prediction.tsv', 'wb') as file:
              file.write(response.content)
```

```
In [137]: image_pred = pd.read_csv('image_prediction.tsv', sep='\t') #loading image_prediction i
```

```
In [138]: image_pred.head() #checking if it is properly read
```

```
Out[138]:              tweet_id                                          jpg_url  \
          0  666020888022790149  https://pbs.twimg.com/media/CT4udn0WwAA0aMy.jpg
          1  666029285002620928  https://pbs.twimg.com/media/CT42GRgUYAA5iDo.jpg
          2  666033412701032449  https://pbs.twimg.com/media/CT4521TWwAEvMyu.jpg
          3  666044226329800704  https://pbs.twimg.com/media/CT5Dr8HUEAA-lEu.jpg
          4  666049248165822465  https://pbs.twimg.com/media/CT5IQmsXIAAKY4A.jpg

             img_num                      p1   p1_conf  p1_dog                  p2  \
          0        1  Welsh_springer_spaniel  0.465074    True              collie
          1        1                 redbone  0.506826    True  miniature_pinscher
          2        1         German_shepherd  0.596461    True            malinois
          3        1     Rhodesian_ridgeback  0.408143    True             redbone
          4        1      miniature_pinscher  0.560311    True          Rottweiler

              p2_conf  p2_dog                   p3   p3_conf  p3_dog
          0  0.156665    True    Shetland_sheepdog  0.061428    True
          1  0.074192    True  Rhodesian_ridgeback  0.072010    True
          2  0.138584    True           bloodhound  0.116197    True
          3  0.360687    True   miniature_pinscher  0.222752    True
          4  0.243682    True             Doberman  0.154629    True
```

3. Use the Tweepy library to query additional data via the Twitter API (tweet_json.txt)

```
In [139]: #reading tweet html with a context manager line by line
          tweets_list = []
```

```
with open('tweet-json.txt', 'r') as file:
    for line in file:
        data = json.loads(line)
        tweets_list.append(data)
```

In [140]:
```
#reading tweets into pandas
tweets_df = pd.DataFrame(tweets_list, columns=['id', 'retweet_count', 'favorite_count'
```

In [141]:
```
#checking if it is read
tweets_df.head()
```

Out[141]:
```
                   id  retweet_count  favorite_count
0  892420643555336193           8853           39467
1  892177421306343426           6514           33819
2  891815181378084864           4328           25461
3  891689557279858688           8964           42908
4  891327558926688256           9774           41048
```

1.3 Assessing Data

In this section, detect and document at least **eight (8) quality issues and two (2) tidiness issue**. You must use **both** visual assessment programmatic assessement to assess the data.

Note: pay attention to the following key points when you access the data.

- You only want original ratings (no retweets) that have images. Though there are 5000+ tweets in the dataset, not all are dog ratings and some are retweets.
- Assessing and cleaning the entire dataset completely would require a lot of time, and is not necessary to practice and demonstrate your skills in data wrangling. Therefore, the requirements of this project are only to assess and clean at least 8 quality issues and at least 2 tidiness issues in this dataset.
- The fact that the rating numerators are greater than the denominators does not need to be cleaned. This unique rating system is a big part of the popularity of WeRateDogs.
- You do not need to gather the tweets beyond August 1st, 2017. You can, but note that you won't be able to gather the image predictions for these tweets since you don't have access to the algorithm used.

1.4 Assessing twitter_archive

In [142]:
```
df_archive.head()
```

Out[142]:
```
             tweet_id  in_reply_to_status_id  in_reply_to_user_id  \
0  892420643555336193                    NaN                  NaN   
1  892177421306343426                    NaN                  NaN   
2  891815181378084864                    NaN                  NaN   
3  891689557279858688                    NaN                  NaN   
4  891327558926688256                    NaN                  NaN   

                   timestamp  \
```

```
0  2017-08-01 16:23:56 +0000
1  2017-08-01 00:17:27 +0000
2  2017-07-31 00:18:03 +0000
3  2017-07-30 15:58:51 +0000
4  2017-07-29 16:00:24 +0000

                                              source  \
0  <a href="http://twitter.com/download/iphone" r...
1  <a href="http://twitter.com/download/iphone" r...
2  <a href="http://twitter.com/download/iphone" r...
3  <a href="http://twitter.com/download/iphone" r...
4  <a href="http://twitter.com/download/iphone" r...

                                                text  retweeted_status_id  \
0  This is Phineas. He's a mystical boy. Only eve...                  NaN
1  This is Tilly. She's just checking pup on you...                   NaN
2  This is Archie. He is a rare Norwegian Pouncin...                  NaN
3  This is Darla. She commenced a snooze mid meal...                  NaN
4  This is Franklin. He would like you to stop ca...                  NaN

   retweeted_status_user_id retweeted_status_timestamp  \
0                       NaN                        NaN
1                       NaN                        NaN
2                       NaN                        NaN
3                       NaN                        NaN
4                       NaN                        NaN

                                       expanded_urls  rating_numerator  \
0  https://twitter.com/dog_rates/status/892420643...                13
1  https://twitter.com/dog_rates/status/892177421...                13
2  https://twitter.com/dog_rates/status/891815181...                12
3  https://twitter.com/dog_rates/status/891689557...                13
4  https://twitter.com/dog_rates/status/891327558...                12

   rating_denominator      name doggo floofer pupper puppo
0                  10   Phineas  None    None   None  None
1                  10     Tilly  None    None   None  None
2                  10    Archie  None    None   None  None
3                  10     Darla  None    None   None  None
4                  10  Franklin  None    None   None  None
```

In [143]:
```
df_archive.sample(5) # checking twitter archive sampling 5 at a time
```

Out[143]:
```
                tweet_id  in_reply_to_status_id  in_reply_to_user_id  \
1127  729838605770891264           7.291135e+17         4.196984e+09
1213  715220193576927233                    NaN                  NaN
741   780496263422808064                    NaN                  NaN
2139  670037189829525505                    NaN                  NaN
```

```
281   839290600511926273                   NaN                 NaN

                     timestamp  \
1127  2016-05-10 01:00:58 +0000
1213  2016-03-30 16:52:36 +0000
741   2016-09-26 19:56:24 +0000
2139  2015-11-27 00:31:29 +0000
281   2017-03-08 01:44:07 +0000

                                                 source  \
1127  <a href="http://twitter.com/download/iphone" r...
1213  <a href="http://twitter.com/download/iphone" r...
741   <a href="http://twitter.com/download/iphone" r...
2139  <a href="http://twitter.com/download/iphone" r...
281   <a href="http://twitter.com/download/iphone" r...

                                                   text  retweeted_status_id  \
1127  "Challenge completed" \n(pupgraded to 12/10) h...                  NaN
1213  This is Nico. His selfie game is strong af. Ex...                  NaN
741   RT @dog_rates: This is Bell. She likes holding...         7.424232e+17
2139  Awesome dog here. Not sure where it is tho. Sp...                  NaN
281   RT @alexmartindawg: THE DRINK IS DR. PUPPER 10...         8.392899e+17

      retweeted_status_user_id retweeted_status_timestamp  \
1127                       NaN                        NaN
1213                       NaN                        NaN
741               4.196984e+09  2016-06-13 18:27:32 +0000
2139                       NaN                        NaN
281               4.119842e+07  2017-03-08 01:41:24 +0000

                                          expanded_urls  rating_numerator  \
1127  https://twitter.com/dog_rates/status/729838605...                12
1213  https://twitter.com/dog_rates/status/715220193...                10
741   https://twitter.com/dog_rates/status/742423170...                12
2139  https://twitter.com/dog_rates/status/670037189...                 5
281   https://twitter.com/alexmartindawg/status/8392...                10

      rating_denominator  name doggo floofer  pupper puppo
1127                  10  None  None    None    None  None
1213                  10  Nico  None    None    None  None
741                   10  Bell  None    None    None  None
2139                  10  None  None    None    None  None
281                   10  None  None    None  pupper  None
```

In [144]: df_archive.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2356 entries, 0 to 2355
```

```
Data columns (total 17 columns):
tweet_id                      2356 non-null int64
in_reply_to_status_id         78 non-null float64
in_reply_to_user_id           78 non-null float64
timestamp                     2356 non-null object
source                        2356 non-null object
text                          2356 non-null object
retweeted_status_id           181 non-null float64
retweeted_status_user_id      181 non-null float64
retweeted_status_timestamp    181 non-null object
expanded_urls                 2297 non-null object
rating_numerator              2356 non-null int64
rating_denominator            2356 non-null int64
name                          2356 non-null object
doggo                         2356 non-null object
floofer                       2356 non-null object
pupper                        2356 non-null object
puppo                         2356 non-null object
dtypes: float64(4), int64(3), object(10)
memory usage: 313.0+ KB
```

In [145]:
```python
# checking if all values in retweeted_statuts_timestamp are all null
df_archive[df_archive['retweeted_status_timestamp'].notnull()]
```

Out[145]:

	tweet_id	in_reply_to_status_id	in_reply_to_user_id	\
19	888202515573088257	NaN	NaN	
32	886054160059072513	NaN	NaN	
36	885311592912609280	NaN	NaN	
68	879130579576475649	NaN	NaN	
73	878404777348136964	NaN	NaN	
74	878316110768087041	NaN	NaN	
78	877611172832227328	NaN	NaN	
91	874434818259525634	NaN	NaN	
95	873697596434513921	NaN	NaN	
97	873337748698140672	NaN	NaN	
101	872668790621863937	NaN	NaN	
109	871166179821445120	NaN	NaN	
118	869988702071779329	NaN	NaN	
124	868639477480148993	NaN	NaN	
130	867072653475098625	NaN	NaN	
132	866816280283807744	NaN	NaN	
137	866094527597207552	NaN	NaN	
146	863471782782697472	NaN	NaN	
155	861769973181624320	NaN	NaN	
159	860981674716409858	NaN	NaN	
160	860924035999428608	NaN	NaN	
165	860177593139703809	NaN	NaN	

```
171   858860390427611136                   NaN                NaN
180   857062103051644929                   NaN                NaN
182   856602993587888130                   NaN                NaN
185   856330835276025856                   NaN                NaN
194   855245323840757760                   NaN                NaN
195   855138241867124737                   NaN                NaN
204   852936405516943360                   NaN                NaN
211   851953902622658560                   NaN                NaN
...                  ...                   ...                ...
784   775096608509886464                   NaN                NaN
794   773336787167145985                   NaN                NaN
800   772615324260794368                   NaN                NaN
811   771171053431250945                   NaN                NaN
815   771004394259247104                   NaN                NaN
818   770743923962707968                   NaN                NaN
822   770093767776997377                   NaN                NaN
826   769335591808995329                   NaN                NaN
829   768909767477751808                   NaN                NaN
833   768554158521745409                   NaN                NaN
841   766864461642756096                   NaN                NaN
847   766078092750233600                   NaN                NaN
860   763167063695355904                   NaN                NaN
868   761750502866649088                   NaN                NaN
872   761371037149827077                   NaN                NaN
885   760153949710192640                   NaN                NaN
890   759566828574212096                   NaN                NaN
895   759159934323924993                   NaN                NaN
908   757729163776290825                   NaN                NaN
911   757597904299253760                   NaN                NaN
926   754874841593970688                   NaN                NaN
937   753298634498793472                   NaN                NaN
943   752701944171524096                   NaN                NaN
949   752309394570878976                   NaN                NaN
1012  747242308580548608                   NaN                NaN
1023  746521445350707200                   NaN                NaN
1043  743835915802583040                   NaN                NaN
1242  711998809858043904                   NaN                NaN
2259  667550904950915073                   NaN                NaN
2260  667550882905632768                   NaN                NaN

                     timestamp  \
19    2017-07-21 01:02:36 +0000
32    2017-07-15 02:45:48 +0000
36    2017-07-13 01:35:06 +0000
68    2017-06-26 00:13:58 +0000
73    2017-06-24 00:09:53 +0000
74    2017-06-23 18:17:33 +0000
78    2017-06-21 19:36:23 +0000
```

```
91    2017-06-13 01:14:41 +0000
95    2017-06-11 00:25:14 +0000
97    2017-06-10 00:35:19 +0000
101   2017-06-08 04:17:07 +0000
109   2017-06-04 00:46:17 +0000
118   2017-05-31 18:47:24 +0000
124   2017-05-28 01:26:04 +0000
130   2017-05-23 17:40:04 +0000
132   2017-05-23 00:41:20 +0000
137   2017-05-21 00:53:21 +0000
146   2017-05-13 19:11:30 +0000
155   2017-05-09 02:29:07 +0000
159   2017-05-06 22:16:42 +0000
160   2017-05-06 18:27:40 +0000
165   2017-05-04 17:01:34 +0000
171   2017-05-01 01:47:28 +0000
180   2017-04-26 02:41:43 +0000
182   2017-04-24 20:17:23 +0000
185   2017-04-24 02:15:55 +0000
194   2017-04-21 02:22:29 +0000
195   2017-04-20 19:16:59 +0000
204   2017-04-14 17:27:40 +0000
211   2017-04-12 00:23:33 +0000
...                         ...
784   2016-09-11 22:20:06 +0000
794   2016-09-07 01:47:12 +0000
800   2016-09-05 02:00:22 +0000
811   2016-09-01 02:21:21 +0000
815   2016-08-31 15:19:06 +0000
818   2016-08-30 22:04:05 +0000
822   2016-08-29 03:00:36 +0000
826   2016-08-27 00:47:53 +0000
829   2016-08-25 20:35:48 +0000
833   2016-08-24 21:02:45 +0000
841   2016-08-20 05:08:29 +0000
847   2016-08-18 01:03:45 +0000
860   2016-08-10 00:16:21 +0000
868   2016-08-06 02:27:27 +0000
872   2016-08-05 01:19:35 +0000
885   2016-08-01 16:43:19 +0000
890   2016-07-31 01:50:18 +0000
895   2016-07-29 22:53:27 +0000
908   2016-07-26 00:08:05 +0000
911   2016-07-25 15:26:30 +0000
926   2016-07-18 03:06:01 +0000
937   2016-07-13 18:42:44 +0000
943   2016-07-12 03:11:42 +0000
949   2016-07-11 01:11:51 +0000
```

```
1012  2016-06-27 01:37:04 +0000
1023  2016-06-25 01:52:36 +0000
1043  2016-06-17 16:01:16 +0000
1242  2016-03-21 19:31:59 +0000
2259  2015-11-20 03:51:52 +0000
2260  2015-11-20 03:51:47 +0000

                                                 source  \
19    <a href="http://twitter.com/download/iphone" r...
32    <a href="http://twitter.com/download/iphone" r...
36    <a href="http://twitter.com/download/iphone" r...
68    <a href="http://twitter.com/download/iphone" r...
73    <a href="http://twitter.com/download/iphone" r...
74    <a href="http://twitter.com/download/iphone" r...
78    <a href="http://twitter.com/download/iphone" r...
91    <a href="http://twitter.com/download/iphone" r...
95    <a href="http://twitter.com/download/iphone" r...
97    <a href="http://twitter.com/download/iphone" r...
101   <a href="http://twitter.com/download/iphone" r...
109   <a href="http://twitter.com/download/iphone" r...
118   <a href="http://twitter.com/download/iphone" r...
124   <a href="http://twitter.com/download/iphone" r...
130   <a href="http://twitter.com/download/iphone" r...
132   <a href="http://twitter.com/download/iphone" r...
137   <a href="http://twitter.com/download/iphone" r...
146   <a href="http://twitter.com/download/iphone" r...
155   <a href="http://twitter.com/download/iphone" r...
159   <a href="http://twitter.com/download/iphone" r...
160   <a href="http://twitter.com/download/iphone" r...
165   <a href="http://twitter.com/download/iphone" r...
171   <a href="http://twitter.com/download/iphone" r...
180   <a href="http://twitter.com/download/iphone" r...
182   <a href="http://twitter.com/download/iphone" r...
185   <a href="http://twitter.com/download/iphone" r...
194   <a href="http://twitter.com/download/iphone" r...
195   <a href="http://twitter.com/download/iphone" r...
204   <a href="http://twitter.com/download/iphone" r...
211   <a href="http://twitter.com/download/iphone" r...
...                                                 ...
784   <a href="http://twitter.com/download/iphone" r...
794   <a href="http://twitter.com/download/iphone" r...
800   <a href="http://twitter.com/download/iphone" r...
811   <a href="http://twitter.com/download/iphone" r...
815   <a href="http://twitter.com/download/iphone" r...
818   <a href="http://twitter.com/download/iphone" r...
822   <a href="http://twitter.com/download/iphone" r...
826   <a href="http://twitter.com/download/iphone" r...
829   <a href="http://twitter.com/download/iphone" r...
```

```
833   <a href="http://twitter.com/download/iphone" r...
841   <a href="http://twitter.com/download/iphone" r...
847   <a href="http://twitter.com/download/iphone" r...
860   <a href="http://twitter.com/download/iphone" r...
868   <a href="http://twitter.com/download/iphone" r...
872   <a href="http://twitter.com/download/iphone" r...
885   <a href="http://twitter.com/download/iphone" r...
890   <a href="http://twitter.com/download/iphone" r...
895   <a href="http://twitter.com/download/iphone" r...
908   <a href="http://twitter.com/download/iphone" r...
911   <a href="http://twitter.com/download/iphone" r...
926   <a href="http://twitter.com/download/iphone" r...
937   <a href="http://twitter.com/download/iphone" r...
943   <a href="http://twitter.com/download/iphone" r...
949   <a href="http://twitter.com/download/iphone" r...
1012  <a href="http://twitter.com/download/iphone" r...
1023  <a href="http://twitter.com/download/iphone" r...
1043  <a href="http://twitter.com/download/iphone" r...
1242  <a href="http://twitter.com/download/iphone" r...
2259  <a href="http://twitter.com" rel="nofollow">Tw...
2260  <a href="http://twitter.com" rel="nofollow">Tw...

                                                   text  retweeted_status_id  \
19    RT @dog_rates: This is Canela. She attempted s...         8.874740e+17
32    RT @Athletics: 12/10 #BATP https://t.co/WxwJmv...         8.860537e+17
36    RT @dog_rates: This is Lilly. She just paralle...         8.305833e+17
68    RT @dog_rates: This is Emmy. She was adopted t...         8.780576e+17
73    RT @dog_rates: Meet Shadow. In an attempt to r...         8.782815e+17
74    RT @dog_rates: Meet Terrance. He's being yelle...         6.690004e+17
78    RT @rachel2195: @dog_rates the boyfriend and h...         8.768508e+17
91    RT @dog_rates: This is Coco. At first I though...         8.663350e+17
95    RT @dog_rates: This is Walter. He won't start ...         8.688804e+17
97    RT @dog_rates: This is Sierra. She's one preci...         8.732138e+17
101   RT @loganamnosis: Penelope here is doing me qu...         8.726576e+17
109   RT @dog_rates: This is Dawn. She's just checki...         8.410770e+17
118   RT @dog_rates: We only rate dogs. This is quit...         8.591970e+17
124   RT @dog_rates: Say hello to Cooper. His expres...         8.685523e+17
130   RT @rachaeleasler: these @dog_rates hats are 1...         8.650134e+17
132   RT @dog_rates: This is Jamesy. He gives a kiss...         8.664507e+17
137   RT @dog_rates: Here's a pupper before and afte...         8.378202e+17
146   RT @dog_rates: Say hello to Quinn. She's quite...         8.630625e+17
155   RT @dog_rates: "Good afternoon class today we'...         8.066291e+17
159   RT @dog_rates: Meet Lorenzo. He's an avid nift...         8.605638e+17
160   RT @tallylott: h*ckin adorable promposal. 13/1...         8.609145e+17
165   RT @dog_rates: Ohboyohboyohboyohboyohboyohboyo...         7.616730e+17
171   RT @dog_rates: Meet Winston. He knows he's a l...         8.395493e+17
180   RT @AaronChewning: First time wearing my @dog_...         8.570611e+17
182   RT @dog_rates: This is Luna. It's her first ti...         8.447048e+17
```

```
185   RT @Jenna_Marbles: @dog_rates Thanks for ratin...        8.563302e+17
194   RT @dog_rates: Meet George. He looks slightly ...        8.421635e+17
195   RT @frasercampbell_: oh my... what's that... b...        8.551225e+17
204   RT @dog_rates: I usually only share these on F...        8.316501e+17
211   RT @dog_rates: This is Astrid. She's a guide d...        8.293743e+17
...                                                ...                  ...
784   RT @dog_rates: After so many requests, this is...        7.403732e+17
794   RT @dog_rates: Meet Fizz. She thinks love is a...        7.713808e+17
800   RT @dog_rates: This is Gromit. He's pupset bec...        7.652221e+17
811   RT @dog_rates: This is Frankie. He's wearing b...        6.733201e+17
815   RT @katieornah: @dog_rates learning a lot at c...        7.710021e+17
818   RT @dog_rates: Here's a doggo blowing bubbles...       7.392382e+17
822   RT @dog_rates: This is just downright precious...        7.410673e+17
826   RT @dog_rates: Ever seen a dog pet another dog...        7.069045e+17
829   RT @dog_rates: When it's Janet from accounting...        7.001438e+17
833   RT @dog_rates: This is Nollie. She's waving at...        7.399792e+17
841   RT @dog_rates: We only rate dogs... this is a ...        7.599238e+17
847   RT @dog_rates: This is Colby. He's currently r...        7.258423e+17
860   RT @dog_rates: Meet Eve. She's a raging alcoho...        6.732953e+17
868   RT @dog_rates: "Tristan do not speak to me wit...        6.853251e+17
872   RT @dog_rates: Oh. My. God. 13/10 magical af h...        7.116948e+17
885   RT @hownottodraw: The story/person behind @dog...        7.601538e+17
890   RT @dog_rates: This... is a Tyrannosaurus rex...       7.395441e+17
895   RT @dog_rates: AT DAWN...\nWE RIDE\n\n11/10 ht...        6.703191e+17
908   RT @dog_rates: This is Chompsky. He lives up t...        6.790626e+17
911   RT @jon_hill987: @dog_rates There is a cunning...        7.575971e+17
926   RT @dog_rates: This is Rubio. He has too much ...        6.791584e+17
937   RT @dog_rates: This is Carly. She's actually 2...        6.815232e+17
943   RT @dog_rates: HEY PUP WHAT'S THE PART OF THE ...        6.835159e+17
949   RT @dog_rates: Everyone needs to watch this. 1...        6.753544e+17
1012  RT @dog_rates: This pupper killed this great w...        7.047611e+17
1023  RT @dog_rates: This is Shaggy. He knows exactl...        6.678667e+17
1043  RT @dog_rates: Extremely intelligent dog here...       6.671383e+17
1242  RT @twitter: @dog_rates Awesome Tweet! 12/10. ...        7.119983e+17
2259  RT @dogratingrating: Exceptional talent. Origi...        6.675487e+17
2260  RT @dogratingrating: Unoriginal idea. Blatant ...        6.675484e+17

      retweeted_status_user_id retweeted_status_timestamp  \
19                4.196984e+09  2017-07-19 00:47:34 +0000
32                1.960740e+07  2017-07-15 02:44:07 +0000
36                4.196984e+09  2017-02-12 01:04:29 +0000
68                4.196984e+09  2017-06-23 01:10:23 +0000
73                4.196984e+09  2017-06-23 16:00:04 +0000
74                4.196984e+09  2015-11-24 03:51:38 +0000
78                5.128045e+08  2017-06-19 17:14:49 +0000
91                4.196984e+09  2017-05-21 16:48:45 +0000
95                4.196984e+09  2017-05-28 17:23:24 +0000
97                4.196984e+09  2017-06-09 16:22:42 +0000
```

101	1.547674e+08	2017-06-08 03:32:35 +0000
109	4.196984e+09	2017-03-13 00:02:39 +0000
118	4.196984e+09	2017-05-02 00:04:57 +0000
124	4.196984e+09	2017-05-27 19:39:34 +0000
130	7.874618e+17	2017-05-18 01:17:25 +0000
132	4.196984e+09	2017-05-22 00:28:40 +0000
137	4.196984e+09	2017-03-04 00:21:08 +0000
146	4.196984e+09	2017-05-12 16:05:02 +0000
155	4.196984e+09	2016-12-07 22:38:52 +0000
159	4.196984e+09	2017-05-05 18:36:06 +0000
160	3.638908e+08	2017-05-06 17:49:42 +0000
165	4.196984e+09	2016-08-05 21:19:27 +0000
171	4.196984e+09	2017-03-08 18:52:12 +0000
180	5.870972e+07	2017-04-26 02:37:47 +0000
182	4.196984e+09	2017-03-23 00:18:10 +0000
185	6.669901e+07	2017-04-24 02:13:14 +0000
194	4.196984e+09	2017-03-16 00:00:07 +0000
195	7.475543e+17	2017-04-20 18:14:33 +0000
204	4.196984e+09	2017-02-14 23:43:18 +0000
211	4.196984e+09	2017-02-08 17:00:26 +0000
...	...	...
784	4.196984e+09	2016-06-08 02:41:38 +0000
794	4.196984e+09	2016-09-01 16:14:48 +0000
800	4.196984e+09	2016-08-15 16:22:20 +0000
811	4.196984e+09	2015-12-06 01:56:44 +0000
815	1.732729e+09	2016-08-31 15:10:07 +0000
818	4.196984e+09	2016-06-04 23:31:25 +0000
822	4.196984e+09	2016-06-10 00:39:48 +0000
826	4.196984e+09	2016-03-07 18:09:06 +0000
829	4.196984e+09	2016-02-18 02:24:13 +0000
833	4.196984e+09	2016-06-07 00:36:02 +0000
841	4.196984e+09	2016-08-01 01:28:46 +0000
847	4.196984e+09	2016-04-29 00:21:01 +0000
860	4.196984e+09	2015-12-06 00:17:55 +0000
868	4.196984e+09	2016-01-08 05:00:14 +0000
872	4.196984e+09	2016-03-20 23:23:54 +0000
885	1.950368e+08	2016-08-01 16:42:51 +0000
890	4.196984e+09	2016-06-05 19:47:03 +0000
895	4.196984e+09	2015-11-27 19:11:49 +0000
908	4.196984e+09	2015-12-21 22:15:18 +0000
911	2.804798e+08	2016-07-25 15:23:28 +0000
926	4.196984e+09	2015-12-22 04:35:49 +0000
937	4.196984e+09	2015-12-28 17:12:42 +0000
943	4.196984e+09	2016-01-03 05:11:12 +0000
949	4.196984e+09	2015-12-11 16:40:19 +0000
1012	4.196984e+09	2016-03-01 20:11:59 +0000
1023	4.196984e+09	2015-11-21 00:46:50 +0000
1043	4.196984e+09	2015-11-19 00:32:12 +0000

```
1242            7.832140e+05  2016-03-21 19:29:52 +0000
2259            4.296832e+09  2015-11-20 03:43:06 +0000
2260            4.296832e+09  2015-11-20 03:41:59 +0000

                                         expanded_urls  rating_numerator  \
19    https://twitter.com/dog_rates/status/887473957...                13
32    https://twitter.com/dog_rates/status/886053434...                12
36    https://twitter.com/dog_rates/status/830583320...                13
68    https://twitter.com/dog_rates/status/878057613...                14
73    https://www.gofundme.com/3yd6y1c,https://twitt...                13
74    https://twitter.com/dog_rates/status/669000397...                11
78    https://twitter.com/rachel2195/status/87685077...                14
91    https://twitter.com/dog_rates/status/866334964...                12
95    https://twitter.com/dog_rates/status/868880397...                14
97    https://www.gofundme.com/help-my-baby-sierra-g...                12
101   https://twitter.com/loganamnosis/status/872657...                14
109   https://twitter.com/dog_rates/status/841077006...                12
118   https://twitter.com/dog_rates/status/859196978...                12
124   https://www.gofundme.com/3ti3nps,https://twitt...                12
130   https://twitter.com/rachaeleasler/status/86501...                13
132   https://twitter.com/dog_rates/status/866450705...                13
137   https://twitter.com/dog_rates/status/837820167...                12
146   https://www.gofundme.com/helpquinny,https://tw...                13
155   https://twitter.com/dog_rates/status/806629075...                13
159   https://www.gofundme.com/help-lorenzo-beat-can...                13
160   https://twitter.com/tallylott/status/860914485...                13
165   https://twitter.com/dog_rates/status/761672994...                10
171   https://twitter.com/dog_rates/status/839549326...                12
180   https://twitter.com/AaronChewning/status/85706...                13
182   https://twitter.com/dog_rates/status/844704788...                13
185                                                 NaN                14
194   https://twitter.com/dog_rates/status/842163532...                12
195   https://twitter.com/frasercampbell_/status/855...                14
204   http://www.gofundme.com/bluethewhitehusky,http...                13
211   https://twitter.com/dog_rates/status/829374341...                13
...                                                 ...               ...
784   https://twitter.com/dog_rates/status/740373189...                 9
794   https://twitter.com/dog_rates/status/771380798...                11
800   https://twitter.com/dog_rates/status/765222098...                10
811   https://twitter.com/dog_rates/status/673320132...                11
815   https://twitter.com/katieornah/status/77100213...                12
818   https://twitter.com/dog_rates/status/739238157...                13
822   https://twitter.com/dog_rates/status/741067306...                12
826   https://vine.co/v/iXQAm5Lrgrh,https://vine.co/...                13
829   https://twitter.com/dog_rates/status/700143752...                10
833   https://twitter.com/dog_rates/status/739979191...                12
841   https://twitter.com/dog_rates/status/759923798...                10
847   https://twitter.com/dog_rates/status/725842289...                12
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860	https://twitter.com/dog_rates/status/673295268...	8
868	https://twitter.com/dog_rates/status/685325112...	10
872	https://twitter.com/dog_rates/status/711694788...	13
885	https://weratedogs.com/pages/about-us,https://...	11
890	https://twitter.com/dog_rates/status/739544079...	10
895	https://twitter.com/dog_rates/status/670319130...	11
908	https://twitter.com/dog_rates/status/679062614...	11
911	https://twitter.com/jon_hill987/status/7575971...	11
926	https://twitter.com/dog_rates/status/679158373...	11
937	https://twitter.com/dog_rates/status/681523177...	12
943	https://vine.co/v/ibvnzrauFuV,https://vine.co/...	11
949	https://twitter.com/dog_rates/status/675354435...	13
1012	https://twitter.com/dog_rates/status/704761120...	13
1023	https://twitter.com/dog_rates/status/667866724...	10
1043	https://twitter.com/dog_rates/status/667138269...	10
1242	https://twitter.com/twitter/status/71199827977...	12
2259	https://twitter.com/dogratingrating/status/667...	12
2260	https://twitter.com/dogratingrating/status/667...	5

	rating_denominator	name	doggo	floofer	pupper	puppo
19	10	Canela	None	None	None	None
32	10	None	None	None	None	None
36	10	Lilly	None	None	None	None
68	10	Emmy	None	None	None	None
73	10	Shadow	None	None	None	None
74	10	Terrance	None	None	None	None
78	10	None	None	None	pupper	None
91	10	Coco	None	None	None	None
95	10	Walter	None	None	None	None
97	10	Sierra	None	None	pupper	None
101	10	None	None	None	None	None
109	10	Dawn	None	None	None	None
118	10	quite	None	None	None	None
124	10	Cooper	None	None	None	None
130	10	None	None	None	None	None
132	10	Jamesy	None	None	pupper	None
137	10	None	None	None	pupper	None
146	10	Quinn	None	None	None	None
155	10	None	None	None	None	None
159	10	Lorenzo	None	None	None	None
160	10	None	None	None	None	None
165	10	None	None	None	None	None
171	10	Winston	None	None	None	None
180	10	None	None	None	None	None
182	10	Luna	None	None	None	None
185	10	None	None	None	None	None
194	10	George	None	None	None	None
195	10	None	None	None	None	None

```
204                 10      None   None    None    None  None
211                 10    Astrid  doggo    None    None  None
...                ...       ...    ...     ...     ...   ...
784                 11      None   None    None    None  None
794                 10      Fizz   None    None    None  None
800                 10    Gromit   None    None    None  None
811                 10   Frankie   None    None    None  None
815                 10      None   None    None  pupper  None
818                 10      None  doggo    None    None  None
822                 10      just  doggo    None  pupper  None
826                 10      None   None    None    None  None
829                 10      None   None    None  pupper  None
833                 10    Nollie   None    None    None  None
841                 10      None   None    None    None  None
847                 10     Colby   None    None    None  None
860                 10       Eve   None    None  pupper  None
868                 10      None   None    None    None  None
872                 10      None   None    None    None  None
885                 10      None   None    None    None  None
890                 10      None   None    None    None  None
895                 10      None   None    None    None  None
908                 10  Chompsky   None    None    None  None
911                 10      None   None    None  pupper  None
926                 10     Rubio   None    None    None  None
937                 10     Carly   None    None    None  None
943                 10      None   None    None    None  None
949                 10      None   None    None    None  None
1012                10      None   None    None  pupper  None
1023                10    Shaggy   None    None    None  None
1043                10      None   None    None    None  None
1242                10      None   None    None    None  None
2259                10      None   None    None    None  None
2260                10      None   None    None    None  None

[181 rows x 17 columns]
```

In [146]:
```
# checking if all values in retweeted_statuts_timestamp are are null
df_archive[df_archive['retweeted_status_timestamp'].notnull()]
```

Out[146]:
```
                tweet_id  in_reply_to_status_id  in_reply_to_user_id  \
19    888202515573088257                    NaN                  NaN
32    886054160059072513                    NaN                  NaN
36    885311592912609280                    NaN                  NaN
68    879130579576475649                    NaN                  NaN
73    878404777348136964                    NaN                  NaN
74    878316110768087041                    NaN                  NaN
78    877611172832227328                    NaN                  NaN
91    874434818259525634                    NaN                  NaN
```

95	873697596434513921	NaN	NaN
97	873337748698140672	NaN	NaN
101	872668790621863937	NaN	NaN
109	871166179821445120	NaN	NaN
118	869988702071779329	NaN	NaN
124	868639477480148993	NaN	NaN
130	867072653475098625	NaN	NaN
132	866816280283807744	NaN	NaN
137	866094527597207552	NaN	NaN
146	863471782782697472	NaN	NaN
155	861769973181624320	NaN	NaN
159	860981674716409858	NaN	NaN
160	860924035999428608	NaN	NaN
165	860177593139703809	NaN	NaN
171	858860390427611136	NaN	NaN
180	857062103051644929	NaN	NaN
182	856602993587888130	NaN	NaN
185	856330835276025856	NaN	NaN
194	855245323840757760	NaN	NaN
195	855138241867124737	NaN	NaN
204	852936405516943360	NaN	NaN
211	851953902622658560	NaN	NaN
...	...	...	...
784	775096608509886464	NaN	NaN
794	773336787167145985	NaN	NaN
800	772615324260794368	NaN	NaN
811	771171053431250945	NaN	NaN
815	771004394259247104	NaN	NaN
818	770743923962707968	NaN	NaN
822	770093767776997377	NaN	NaN
826	769335591808995329	NaN	NaN
829	768909767477751808	NaN	NaN
833	768554158521745409	NaN	NaN
841	766864461642756096	NaN	NaN
847	766078092750233600	NaN	NaN
860	763167063695355904	NaN	NaN
868	761750502866649088	NaN	NaN
872	761371037149827077	NaN	NaN
885	760153949710192640	NaN	NaN
890	759566828574212096	NaN	NaN
895	759159934323924993	NaN	NaN
908	757729163776290825	NaN	NaN
911	757597904299253760	NaN	NaN
926	754874841593970688	NaN	NaN
937	753298634498793472	NaN	NaN
943	752701944171524096	NaN	NaN
949	752309394570878976	NaN	NaN
1012	747242308580548608	NaN	NaN

```
1023  746521445350707200                    NaN                    NaN
1043  743835915802583040                    NaN                    NaN
1242  711998809858043904                    NaN                    NaN
2259  667550904950915073                    NaN                    NaN
2260  667550882905632768                    NaN                    NaN

                     timestamp  \
19    2017-07-21 01:02:36 +0000
32    2017-07-15 02:45:48 +0000
36    2017-07-13 01:35:06 +0000
68    2017-06-26 00:13:58 +0000
73    2017-06-24 00:09:53 +0000
74    2017-06-23 18:17:33 +0000
78    2017-06-21 19:36:23 +0000
91    2017-06-13 01:14:41 +0000
95    2017-06-11 00:25:14 +0000
97    2017-06-10 00:35:19 +0000
101   2017-06-08 04:17:07 +0000
109   2017-06-04 00:46:17 +0000
118   2017-05-31 18:47:24 +0000
124   2017-05-28 01:26:04 +0000
130   2017-05-23 17:40:04 +0000
132   2017-05-23 00:41:20 +0000
137   2017-05-21 00:53:21 +0000
146   2017-05-13 19:11:30 +0000
155   2017-05-09 02:29:07 +0000
159   2017-05-06 22:16:42 +0000
160   2017-05-06 18:27:40 +0000
165   2017-05-04 17:01:34 +0000
171   2017-05-01 01:47:28 +0000
180   2017-04-26 02:41:43 +0000
182   2017-04-24 20:17:23 +0000
185   2017-04-24 02:15:55 +0000
194   2017-04-21 02:22:29 +0000
195   2017-04-20 19:16:59 +0000
204   2017-04-14 17:27:40 +0000
211   2017-04-12 00:23:33 +0000
...                         ...
784   2016-09-11 22:20:06 +0000
794   2016-09-07 01:47:12 +0000
800   2016-09-05 02:00:22 +0000
811   2016-09-01 02:21:21 +0000
815   2016-08-31 15:19:06 +0000
818   2016-08-30 22:04:05 +0000
822   2016-08-29 03:00:36 +0000
826   2016-08-27 00:47:53 +0000
829   2016-08-25 20:35:48 +0000
833   2016-08-24 21:02:45 +0000
```

```
841   2016-08-20 05:08:29 +0000
847   2016-08-18 01:03:45 +0000
860   2016-08-10 00:16:21 +0000
868   2016-08-06 02:27:27 +0000
872   2016-08-05 01:19:35 +0000
885   2016-08-01 16:43:19 +0000
890   2016-07-31 01:50:18 +0000
895   2016-07-29 22:53:27 +0000
908   2016-07-26 00:08:05 +0000
911   2016-07-25 15:26:30 +0000
926   2016-07-18 03:06:01 +0000
937   2016-07-13 18:42:44 +0000
943   2016-07-12 03:11:42 +0000
949   2016-07-11 01:11:51 +0000
1012  2016-06-27 01:37:04 +0000
1023  2016-06-25 01:52:36 +0000
1043  2016-06-17 16:01:16 +0000
1242  2016-03-21 19:31:59 +0000
2259  2015-11-20 03:51:52 +0000
2260  2015-11-20 03:51:47 +0000

                                                 source  \
19    <a href="http://twitter.com/download/iphone" r...
32    <a href="http://twitter.com/download/iphone" r...
36    <a href="http://twitter.com/download/iphone" r...
68    <a href="http://twitter.com/download/iphone" r...
73    <a href="http://twitter.com/download/iphone" r...
74    <a href="http://twitter.com/download/iphone" r...
78    <a href="http://twitter.com/download/iphone" r...
91    <a href="http://twitter.com/download/iphone" r...
95    <a href="http://twitter.com/download/iphone" r...
97    <a href="http://twitter.com/download/iphone" r...
101   <a href="http://twitter.com/download/iphone" r...
109   <a href="http://twitter.com/download/iphone" r...
118   <a href="http://twitter.com/download/iphone" r...
124   <a href="http://twitter.com/download/iphone" r...
130   <a href="http://twitter.com/download/iphone" r...
132   <a href="http://twitter.com/download/iphone" r...
137   <a href="http://twitter.com/download/iphone" r...
146   <a href="http://twitter.com/download/iphone" r...
155   <a href="http://twitter.com/download/iphone" r...
159   <a href="http://twitter.com/download/iphone" r...
160   <a href="http://twitter.com/download/iphone" r...
165   <a href="http://twitter.com/download/iphone" r...
171   <a href="http://twitter.com/download/iphone" r...
180   <a href="http://twitter.com/download/iphone" r...
182   <a href="http://twitter.com/download/iphone" r...
185   <a href="http://twitter.com/download/iphone" r...
```

```
194   <a href="http://twitter.com/download/iphone" r...
195   <a href="http://twitter.com/download/iphone" r...
204   <a href="http://twitter.com/download/iphone" r...
211   <a href="http://twitter.com/download/iphone" r...
...                                                ...
784   <a href="http://twitter.com/download/iphone" r...
794   <a href="http://twitter.com/download/iphone" r...
800   <a href="http://twitter.com/download/iphone" r...
811   <a href="http://twitter.com/download/iphone" r...
815   <a href="http://twitter.com/download/iphone" r...
818   <a href="http://twitter.com/download/iphone" r...
822   <a href="http://twitter.com/download/iphone" r...
826   <a href="http://twitter.com/download/iphone" r...
829   <a href="http://twitter.com/download/iphone" r...
833   <a href="http://twitter.com/download/iphone" r...
841   <a href="http://twitter.com/download/iphone" r...
847   <a href="http://twitter.com/download/iphone" r...
860   <a href="http://twitter.com/download/iphone" r...
868   <a href="http://twitter.com/download/iphone" r...
872   <a href="http://twitter.com/download/iphone" r...
885   <a href="http://twitter.com/download/iphone" r...
890   <a href="http://twitter.com/download/iphone" r...
895   <a href="http://twitter.com/download/iphone" r...
908   <a href="http://twitter.com/download/iphone" r...
911   <a href="http://twitter.com/download/iphone" r...
926   <a href="http://twitter.com/download/iphone" r...
937   <a href="http://twitter.com/download/iphone" r...
943   <a href="http://twitter.com/download/iphone" r...
949   <a href="http://twitter.com/download/iphone" r...
1012  <a href="http://twitter.com/download/iphone" r...
1023  <a href="http://twitter.com/download/iphone" r...
1043  <a href="http://twitter.com/download/iphone" r...
1242  <a href="http://twitter.com/download/iphone" r...
2259  <a href="http://twitter.com" rel="nofollow">Tw...
2260  <a href="http://twitter.com" rel="nofollow">Tw...

                                                   text  retweeted_status_id  \
19    RT @dog_rates: This is Canela. She attempted s...         8.874740e+17
32    RT @Athletics: 12/10 #BATP https://t.co/WxwJmv...         8.860537e+17
36    RT @dog_rates: This is Lilly. She just paralle...         8.305833e+17
68    RT @dog_rates: This is Emmy. She was adopted t...         8.780576e+17
73    RT @dog_rates: Meet Shadow. In an attempt to r...         8.782815e+17
74    RT @dog_rates: Meet Terrance. He's being yelle...         6.690004e+17
78    RT @rachel2195: @dog_rates the boyfriend and h...         8.768508e+17
91    RT @dog_rates: This is Coco. At first I though...         8.663350e+17
95    RT @dog_rates: This is Walter. He won't start ...         8.688804e+17
97    RT @dog_rates: This is Sierra. She's one preci...         8.732138e+17
101   RT @loganamnosis: Penelope here is doing me qu...         8.726576e+17
```

109	RT @dog_rates: This is Dawn. She's just checki...	8.410770e+17
118	RT @dog_rates: We only rate dogs. This is quit...	8.591970e+17
124	RT @dog_rates: Say hello to Cooper. His expres...	8.685523e+17
130	RT @rachaeleasler: these @dog_rates hats are 1...	8.650134e+17
132	RT @dog_rates: This is Jamesy. He gives a kiss...	8.664507e+17
137	RT @dog_rates: Here's a pupper before and afte...	8.378202e+17
146	RT @dog_rates: Say hello to Quinn. She's quite...	8.630625e+17
155	RT @dog_rates: "Good afternoon class today we'...	8.066291e+17
159	RT @dog_rates: Meet Lorenzo. He's an avid nift...	8.605638e+17
160	RT @tallylott: h*ckin adorable promposal. 13/1...	8.609145e+17
165	RT @dog_rates: Ohboyohboyohboyohboyohboyohboyo...	7.616730e+17
171	RT @dog_rates: Meet Winston. He knows he's a l...	8.395493e+17
180	RT @AaronChewning: First time wearing my @dog_...	8.570611e+17
182	RT @dog_rates: This is Luna. It's her first ti...	8.447048e+17
185	RT @Jenna_Marbles: @dog_rates Thanks for ratin...	8.563302e+17
194	RT @dog_rates: Meet George. He looks slightly ...	8.421635e+17
195	RT @frasercampbell_: oh my... what's that... b...	8.551225e+17
204	RT @dog_rates: I usually only share these on F...	8.316501e+17
211	RT @dog_rates: This is Astrid. She's a guide d...	8.293743e+17
...	...	...
784	RT @dog_rates: After so many requests, this is...	7.403732e+17
794	RT @dog_rates: Meet Fizz. She thinks love is a...	7.713808e+17
800	RT @dog_rates: This is Gromit. He's pupset bec...	7.652221e+17
811	RT @dog_rates: This is Frankie. He's wearing b...	6.733201e+17
815	RT @katieornah: @dog_rates learning a lot at c...	7.710021e+17
818	RT @dog_rates: Here's a doggo blowing bubbles...	7.392382e+17
822	RT @dog_rates: This is just downright precious...	7.410673e+17
826	RT @dog_rates: Ever seen a dog pet another dog...	7.069045e+17
829	RT @dog_rates: When it's Janet from accounting...	7.001438e+17
833	RT @dog_rates: This is Nollie. She's waving at...	7.399792e+17
841	RT @dog_rates: We only rate dogs... this is a ...	7.599238e+17
847	RT @dog_rates: This is Colby. He's currently r...	7.258423e+17
860	RT @dog_rates: Meet Eve. She's a raging alcoho...	6.732953e+17
868	RT @dog_rates: "Tristan do not speak to me wit...	6.853251e+17
872	RT @dog_rates: Oh. My. God. 13/10 magical af h...	7.116948e+17
885	RT @hownottodraw: The story/person behind @dog...	7.601538e+17
890	RT @dog_rates: This... is a Tyrannosaurus rex...	7.395441e+17
895	RT @dog_rates: AT DAWN...\nWE RIDE\n\n11/10 ht...	6.703191e+17
908	RT @dog_rates: This is Chompsky. He lives up t...	6.790626e+17
911	RT @jon_hill987: @dog_rates There is a cunning...	7.575971e+17
926	RT @dog_rates: This is Rubio. He has too much ...	6.791584e+17
937	RT @dog_rates: This is Carly. She's actually 2...	6.815232e+17
943	RT @dog_rates: HEY PUP WHAT'S THE PART OF THE ...	6.835159e+17
949	RT @dog_rates: Everyone needs to watch this. 1...	6.753544e+17
1012	RT @dog_rates: This pupper killed this great w...	7.047611e+17
1023	RT @dog_rates: This is Shaggy. He knows exactl...	6.678667e+17
1043	RT @dog_rates: Extremely intelligent dog here...	6.671383e+17
1242	RT @twitter: @dog_rates Awesome Tweet! 12/10. ...	7.119983e+17

```
2259  RT @dogratingrating: Exceptional talent. Origi...         6.675487e+17
2260  RT @dogratingrating: Unoriginal idea. Blatant ...         6.675484e+17

      retweeted_status_user_id retweeted_status_timestamp  \
19                4.196984e+09  2017-07-19 00:47:34 +0000
32                1.960740e+07  2017-07-15 02:44:07 +0000
36                4.196984e+09  2017-02-12 01:04:29 +0000
68                4.196984e+09  2017-06-23 01:10:23 +0000
73                4.196984e+09  2017-06-23 16:00:04 +0000
74                4.196984e+09  2015-11-24 03:51:38 +0000
78                5.128045e+08  2017-06-19 17:14:49 +0000
91                4.196984e+09  2017-05-21 16:48:45 +0000
95                4.196984e+09  2017-05-28 17:23:24 +0000
97                4.196984e+09  2017-06-09 16:22:42 +0000
101               1.547674e+08  2017-06-08 03:32:35 +0000
109               4.196984e+09  2017-03-13 00:02:39 +0000
118               4.196984e+09  2017-05-02 00:04:57 +0000
124               4.196984e+09  2017-05-27 19:39:34 +0000
130               7.874618e+17  2017-05-18 01:17:25 +0000
132               4.196984e+09  2017-05-22 00:28:40 +0000
137               4.196984e+09  2017-03-04 00:21:08 +0000
146               4.196984e+09  2017-05-12 16:05:02 +0000
155               4.196984e+09  2016-12-07 22:38:52 +0000
159               4.196984e+09  2017-05-05 18:36:06 +0000
160               3.638908e+08  2017-05-06 17:49:42 +0000
165               4.196984e+09  2016-08-05 21:19:27 +0000
171               4.196984e+09  2017-03-08 18:52:12 +0000
180               5.870972e+07  2017-04-26 02:37:47 +0000
182               4.196984e+09  2017-03-23 00:18:10 +0000
185               6.669901e+07  2017-04-24 02:13:14 +0000
194               4.196984e+09  2017-03-16 00:00:07 +0000
195               7.475543e+17  2017-04-20 18:14:33 +0000
204               4.196984e+09  2017-02-14 23:43:18 +0000
211               4.196984e+09  2017-02-08 17:00:26 +0000
...                        ...                        ...
784               4.196984e+09  2016-06-08 02:41:38 +0000
794               4.196984e+09  2016-09-01 16:14:48 +0000
800               4.196984e+09  2016-08-15 16:22:20 +0000
811               4.196984e+09  2015-12-06 01:56:44 +0000
815               1.732729e+09  2016-08-31 15:10:07 +0000
818               4.196984e+09  2016-06-04 23:31:25 +0000
822               4.196984e+09  2016-06-10 00:39:48 +0000
826               4.196984e+09  2016-03-07 18:09:06 +0000
829               4.196984e+09  2016-02-18 02:24:13 +0000
833               4.196984e+09  2016-06-07 00:36:02 +0000
841               4.196984e+09  2016-08-01 01:28:46 +0000
847               4.196984e+09  2016-04-29 00:21:01 +0000
860               4.196984e+09  2015-12-06 00:17:55 +0000
```

868	4.196984e+09	2016-01-08 05:00:14 +0000
872	4.196984e+09	2016-03-20 23:23:54 +0000
885	1.950368e+08	2016-08-01 16:42:51 +0000
890	4.196984e+09	2016-06-05 19:47:03 +0000
895	4.196984e+09	2015-11-27 19:11:49 +0000
908	4.196984e+09	2015-12-21 22:15:18 +0000
911	2.804798e+08	2016-07-25 15:23:28 +0000
926	4.196984e+09	2015-12-22 04:35:49 +0000
937	4.196984e+09	2015-12-28 17:12:42 +0000
943	4.196984e+09	2016-01-03 05:11:12 +0000
949	4.196984e+09	2015-12-11 16:40:19 +0000
1012	4.196984e+09	2016-03-01 20:11:59 +0000
1023	4.196984e+09	2015-11-21 00:46:50 +0000
1043	4.196984e+09	2015-11-19 00:32:12 +0000
1242	7.832140e+05	2016-03-21 19:29:52 +0000
2259	4.296832e+09	2015-11-20 03:43:06 +0000
2260	4.296832e+09	2015-11-20 03:41:59 +0000

	expanded_urls	rating_numerator \
19	https://twitter.com/dog_rates/status/887473957...	13
32	https://twitter.com/dog_rates/status/886053434...	12
36	https://twitter.com/dog_rates/status/830583320...	13
68	https://twitter.com/dog_rates/status/878057613...	14
73	https://www.gofundme.com/3yd6y1c,https://twitt...	13
74	https://twitter.com/dog_rates/status/669000397...	11
78	https://twitter.com/rachel2195/status/87685077...	14
91	https://twitter.com/dog_rates/status/866334964...	12
95	https://twitter.com/dog_rates/status/868880397...	14
97	https://www.gofundme.com/help-my-baby-sierra-g...	12
101	https://twitter.com/loganamnosis/status/872657...	14
109	https://twitter.com/dog_rates/status/841077006...	12
118	https://twitter.com/dog_rates/status/859196978...	12
124	https://www.gofundme.com/3ti3nps,https://twitt...	12
130	https://twitter.com/rachaeleasler/status/86501...	13
132	https://twitter.com/dog_rates/status/866450705...	13
137	https://twitter.com/dog_rates/status/837820167...	12
146	https://www.gofundme.com/helpquinny,https://tw...	13
155	https://twitter.com/dog_rates/status/806629075...	13
159	https://www.gofundme.com/help-lorenzo-beat-can...	13
160	https://twitter.com/tallylott/status/860914485...	13
165	https://twitter.com/dog_rates/status/761672994...	10
171	https://twitter.com/dog_rates/status/839549326...	12
180	https://twitter.com/AaronChewning/status/85706...	13
182	https://twitter.com/dog_rates/status/844704788...	13
185	NaN	14
194	https://twitter.com/dog_rates/status/842163532...	12
195	https://twitter.com/frasercampbell_/status/855...	14
204	http://www.gofundme.com/bluethewhitehusky,http...	13

```
211   https://twitter.com/dog_rates/status/829374341...                 13
...                                                 ...                ...
784   https://twitter.com/dog_rates/status/740373189...                  9
794   https://twitter.com/dog_rates/status/771380798...                 11
800   https://twitter.com/dog_rates/status/765222098...                 10
811   https://twitter.com/dog_rates/status/673320132...                 11
815   https://twitter.com/katieornah/status/77100213...                 12
818   https://twitter.com/dog_rates/status/739238157...                 13
822   https://twitter.com/dog_rates/status/741067306...                 12
826   https://vine.co/v/iXQAm5Lrgrh,https://vine.co/...                 13
829   https://twitter.com/dog_rates/status/700143752...                 10
833   https://twitter.com/dog_rates/status/739979191...                 12
841   https://twitter.com/dog_rates/status/759923798...                 10
847   https://twitter.com/dog_rates/status/725842289...                 12
860   https://twitter.com/dog_rates/status/673295268...                  8
868   https://twitter.com/dog_rates/status/685325112...                 10
872   https://twitter.com/dog_rates/status/711694788...                 13
885   https://weratedogs.com/pages/about-us,https://...                 11
890   https://twitter.com/dog_rates/status/739544079...                 10
895   https://twitter.com/dog_rates/status/670319130...                 11
908   https://twitter.com/dog_rates/status/679062614...                 11
911   https://twitter.com/jon_hill987/status/7575971...                 11
926   https://twitter.com/dog_rates/status/679158373...                 11
937   https://twitter.com/dog_rates/status/681523177...                 12
943   https://vine.co/v/ibvnzrauFuV,https://vine.co/...                 11
949   https://twitter.com/dog_rates/status/675354435...                 13
1012  https://twitter.com/dog_rates/status/704761120...                 13
1023  https://twitter.com/dog_rates/status/667866724...                 10
1043  https://twitter.com/dog_rates/status/667138269...                 10
1242  https://twitter.com/twitter/status/71199827977...                 12
2259  https://twitter.com/dogratingrating/status/667...                 12
2260  https://twitter.com/dogratingrating/status/667...                  5

      rating_denominator      name  doggo floofer  pupper puppo
19                    10    Canela   None    None    None  None
32                    10      None   None    None    None  None
36                    10     Lilly   None    None    None  None
68                    10      Emmy   None    None    None  None
73                    10    Shadow   None    None    None  None
74                    10  Terrance   None    None    None  None
78                    10      None   None    None  pupper  None
91                    10      Coco   None    None    None  None
95                    10    Walter   None    None    None  None
97                    10    Sierra   None    None  pupper  None
101                   10      None   None    None    None  None
109                   10      Dawn   None    None    None  None
118                   10     quite   None    None    None  None
124                   10    Cooper   None    None    None  None
```

130	10	None	None	None	None	None
132	10	Jamesy	None	None	pupper	None
137	10	None	None	None	pupper	None
146	10	Quinn	None	None	None	None
155	10	None	None	None	None	None
159	10	Lorenzo	None	None	None	None
160	10	None	None	None	None	None
165	10	None	None	None	None	None
171	10	Winston	None	None	None	None
180	10	None	None	None	None	None
182	10	Luna	None	None	None	None
185	10	None	None	None	None	None
194	10	George	None	None	None	None
195	10	None	None	None	None	None
204	10	None	None	None	None	None
211	10	Astrid	doggo	None	None	None
...	...	...	...	...	...	...
784	11	None	None	None	None	None
794	10	Fizz	None	None	None	None
800	10	Gromit	None	None	None	None
811	10	Frankie	None	None	None	None
815	10	None	None	None	pupper	None
818	10	None	doggo	None	None	None
822	10	just	doggo	None	pupper	None
826	10	None	None	None	None	None
829	10	None	None	None	pupper	None
833	10	Nollie	None	None	None	None
841	10	None	None	None	None	None
847	10	Colby	None	None	None	None
860	10	Eve	None	None	pupper	None
868	10	None	None	None	None	None
872	10	None	None	None	None	None
885	10	None	None	None	None	None
890	10	None	None	None	None	None
895	10	None	None	None	None	None
908	10	Chompsky	None	None	None	None
911	10	None	None	None	pupper	None
926	10	Rubio	None	None	None	None
937	10	Carly	None	None	None	None
943	10	None	None	None	None	None
949	10	None	None	None	None	None
1012	10	None	None	None	pupper	None
1023	10	Shaggy	None	None	None	None
1043	10	None	None	None	None	None
1242	10	None	None	None	None	None
2259	10	None	None	None	None	None
2260	10	None	None	None	None	None

```
[181 rows x 17 columns]
```

In [147]:
```
df_archive.sample(5)
```

Out[147]:
```
                tweet_id  in_reply_to_status_id  in_reply_to_user_id  \
1248  711363825979756544                    NaN                  NaN
1002  747885874273214464                    NaN                  NaN
1966  673343217010679808                    NaN                  NaN
2252  667801013445750784                    NaN                  NaN
1104  735137028879360001                    NaN                  NaN

                      timestamp  \
1248  2016-03-20 01:28:47 +0000
1002  2016-06-28 20:14:22 +0000
1966  2015-12-06 03:28:27 +0000
2252  2015-11-20 20:25:43 +0000
1104  2016-05-24 15:55:00 +0000

                                                 source  \
1248  <a href="http://twitter.com/download/iphone" r...
1002  <a href="http://twitter.com/download/iphone" r...
1966  <a href="http://twitter.com/download/iphone" r...
2252  <a href="http://twitter.com/download/iphone" r...
1104  <a href="http://twitter.com/download/iphone" r...

                                                   text  retweeted_status_id  \
1248  "Please, no puparazzi" 11/10 https://t.co/nJIX...                  NaN
1002  This is a mighty rare blue-tailed hammer sherk...                  NaN
1966  IT'S SO SMALL ERMERGERF 11/10 https://t.co/dNU...                  NaN
2252               OMIGOD 12/10 https://t.co/SVMF4Frf1w                  NaN
1104  Meet Buckley. His family &amp; some neighbors ...                  NaN

      retweeted_status_user_id retweeted_status_timestamp  \
1248                       NaN                        NaN
1002                       NaN                        NaN
1966                       NaN                        NaN
2252                       NaN                        NaN
1104                       NaN                        NaN

                                          expanded_urls  rating_numerator  \
1248  https://twitter.com/dog_rates/status/711363825...                11
1002  https://twitter.com/dog_rates/status/747885874...                 8
1966  https://twitter.com/dog_rates/status/673343217...                11
2252  https://twitter.com/dog_rates/status/667801013...                12
1104  https://twitter.com/dog_rates/status/735137028...                 9

      rating_denominator     name doggo floofer  pupper puppo
1248                  10     None  None    None    None  None
```

```
1002                  10        a  None    None    None  None
1966                  10     None  None    None    None  None
2252                  10     None  None    None    None  None
1104                  10  Buckley  None    None  pupper  None
```

```
In [148]: df_archive['text'][59] # checking text at index 59. tweet not of dog
```

```
Out[148]: "Ugh not again. We only rate dogs. Please don't send in well-dressed  floppy-tongued s
```

```
In [149]: df_archive.columns
```

```
Out[149]: Index(['tweet_id', 'in_reply_to_status_id', 'in_reply_to_user_id', 'timestamp',
                 'source', 'text', 'retweeted_status_id', 'retweeted_status_user_id',
                 'retweeted_status_timestamp', 'expanded_urls', 'rating_numerator',
                 'rating_denominator', 'name', 'doggo', 'floofer', 'pupper', 'puppo'],
                dtype='object')
```

```
In [150]: # checking values in dog stages; doggo, floofer, pupper and puppo
          df_archive[['doggo', 'floofer', 'pupper', 'puppo']]
```

```
Out[150]:       doggo floofer  pupper  puppo
          0      None    None    None   None
          1      None    None    None   None
          2      None    None    None   None
          3      None    None    None   None
          4      None    None    None   None
          5      None    None    None   None
          6      None    None    None   None
          7      None    None    None   None
          8      None    None    None   None
          9     doggo    None    None   None
          10     None    None    None   None
          11     None    None    None   None
          12     None    None    None  puppo
          13     None    None    None   None
          14     None    None    None  puppo
          15     None    None    None   None
          16     None    None    None   None
          17     None    None    None   None
          18     None    None    None   None
          19     None    None    None   None
          20     None    None    None   None
          21     None    None    None   None
          22     None    None    None   None
          23     None    None    None   None
          24     None    None    None   None
          25     None    None    None   None
          26     None    None    None   None
          27     None    None    None   None
```

```
28      None    None     None    None
29      None    None   pupper    None
...      ...     ...      ...     ...
2326    None    None     None    None
2327    None    None     None    None
2328    None    None     None    None
2329    None    None     None    None
2330    None    None     None    None
2331    None    None     None    None
2332    None    None     None    None
2333    None    None     None    None
2334    None    None     None    None
2335    None    None     None    None
2336    None    None     None    None
2337    None    None     None    None
2338    None    None     None    None
2339    None    None     None    None
2340    None    None     None    None
2341    None    None     None    None
2342    None    None     None    None
2343    None    None     None    None
2344    None    None     None    None
2345    None    None     None    None
2346    None    None     None    None
2347    None    None     None    None
2348    None    None     None    None
2349    None    None     None    None
2350    None    None     None    None
2351    None    None     None    None
2352    None    None     None    None
2353    None    None     None    None
2354    None    None     None    None
2355    None    None     None    None

[2356 rows x 4 columns]
```

In [257]:
```
# checking for duplicated entries
df_archive.duplicated().sum()
```

Out[257]:
```
0
```

1.4.1 Quality issues with twitter archive/ df_archive

1. issues with extraction of ratings
2. wrong datatype: times stamp needs to be changed from object to datetime
3. some the tweets are not dog eg index 59 and 103
4. inconsistency values in dog stages [doggo, floofer, pupper, puppo, None]

5. source not properly extracted
6. missing value in in_reply_to_status_id
7. missing value in in_reply_to_user_id
8. missing value in in_reply_to_status_id
9. missing value in retweeted_status_id
10. missing value in retweeted_status_user_id
11. missing vslue in retweeted_status_timestamp

1.4.2 Tidiness issues tweet_archive

1. - dog stage should be one column in twitter_archive

1.5 Assessing Image Prediction(image_pred)

In [151]:
```
image_pred.head()
```

Out[151]:
```
             tweet_id                                          jpg_url  \
0  666020888022790149  https://pbs.twimg.com/media/CT4udn0WwAA0aMy.jpg
1  666029285002620928  https://pbs.twimg.com/media/CT42GRgUYAA5iDo.jpg
2  666033412701032449  https://pbs.twimg.com/media/CT4521TWwAEvMyu.jpg
3  666044226329800704  https://pbs.twimg.com/media/CT5Dr8HUEAA-lEu.jpg
4  666049248165822465  https://pbs.twimg.com/media/CT5IQmsXIAAKY4A.jpg

   img_num                      p1   p1_conf  p1_dog                  p2  \
0        1  Welsh_springer_spaniel  0.465074    True              collie
1        1                 redbone  0.506826    True  miniature_pinscher
2        1         German_shepherd  0.596461    True            malinois
3        1     Rhodesian_ridgeback  0.408143    True             redbone
4        1      miniature_pinscher  0.560311    True          Rottweiler

    p2_conf  p2_dog                   p3   p3_conf  p3_dog
0  0.156665    True    Shetland_sheepdog  0.061428    True
1  0.074192    True  Rhodesian_ridgeback  0.072010    True
2  0.138584    True           bloodhound  0.116197    True
3  0.360687    True   miniature_pinscher  0.222752    True
4  0.243682    True             Doberman  0.154629    True
```

In [152]:
```
image_pred.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2075 entries, 0 to 2074
Data columns (total 12 columns):
tweet_id    2075 non-null int64
jpg_url     2075 non-null object
img_num     2075 non-null int64
```

```
p1          2075 non-null object
p1_conf     2075 non-null float64
p1_dog      2075 non-null bool
p2          2075 non-null object
p2_conf     2075 non-null float64
p2_dog      2075 non-null bool
p3          2075 non-null object
p3_conf     2075 non-null float64
p3_dog      2075 non-null bool
dtypes: bool(3), float64(3), int64(2), object(4)
memory usage: 152.1+ KB
```

In [153]:
```
image_pred.sample(5)
```

Out[153]:
```
                tweet_id                                          jpg_url  \
481   675362609739206656  https://pbs.twimg.com/media/CV9etctWUAAl5Hp.jpg
1022  710272297844797440  https://pbs.twimg.com/media/Cdtk414WoAIUG0v.jpg
538   676949632774234114  https://pbs.twimg.com/media/CWUCGMtWEAAjXnS.jpg
762   688898160958271489  https://pbs.twimg.com/media/CY91OENWUAE5agj.jpg
999   708479650088034305  https://pbs.twimg.com/media/CdUGcLMWAAI42q0.jpg

      img_num                      p1   p1_conf  p1_dog  \
481         1      Labrador_retriever  0.479008    True
1022        1      Old_English_sheepdog  0.586307    True
538         1  Welsh_springer_spaniel  0.206479    True
762         1            Ibizan_hound  0.853170    True
999         1                Shih-Tzu  0.218479    True

                           p2   p2_conf  p2_dog                p3   p3_conf  \
481                  ice_bear  0.218289   False            kuvasz  0.139911
1022  wire-haired_fox_terrier  0.118622    True  Lakeland_terrier  0.106806
538             Saint_Bernard  0.139339    True             boxer  0.114606
762                 Chihuahua  0.039897    True  Italian_greyhound  0.035220
999                     Lhasa  0.201966    True   Norfolk_terrier  0.165225

      p3_dog
481     True
1022    True
538     True
762     True
999     True
```

In [154]:
```
image_pred.iloc[53,1] # extracting the url of an image not dog
```

Out[154]:
```
'https://pbs.twimg.com/media/CUG0bC0U8AAw2su.jpg'
```

In [155]:
```
# displaying an image not dog using url extracted above
from IPython.display import Image
Image(url= "https://pbs.twimg.com/media/CUG0bC0U8AAw2su.jpg ", width=400, height=400)
```

Out[155]: <IPython.core.display.Image object>

In [258]:
```
# checking for duplicated values
image_pred.duplicated().sum()
```

Out[258]: 0

1.5.1 Quality issue

1. some images are not dog images

1.5.2 Tidiness issue

1. dog prediction and confidence should be one column

1.6 Assessing tweets_df

In [156]:
```
tweets_df.head()
```

Out[156]:

	id	retweet_count	favorite_count
0	892420643555336193	8853	39467
1	892177421306343426	6514	33819
2	891815181378084864	4328	25461
3	891689557279858688	8964	42908
4	891327558926688256	9774	41048

In [157]:
```
tweets_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2354 entries, 0 to 2353
Data columns (total 3 columns):
id                2354 non-null int64
retweet_count     2354 non-null int64
favorite_count    2354 non-null int64
dtypes: int64(3)
memory usage: 55.2 KB
```

In [158]:
```
tweets_df.sample(5)
```

Out[158]:

	id	retweet_count	favorite_count
1288	708130923141795840	943	3707
98	872967104147763200	5669	28031
506	812709060537683968	1665	7373
981	749395845976588288	3951	9488
245	845677943972139009	5365	27154

In [260]:
```
# checking for duplicated entries
tweets_df.duplicated().sum()
```

Out[260]: 0

1.7 Quality Issue with tweets_df

- none

1.8 Tidyness issue

- change id to tweet_id for merging

1.9 Cleaning Data

In this section, clean **all** of the issues you documented while assessing.

Note: Make a copy of the original data before cleaning. Cleaning includes merging individual pieces of data according to the rules of tidy data. The result should be a high-quality and tidy master pandas DataFrame (or DataFrames, if appropriate).

```
In [159]: # Make copies of original three of dataframes
          df_archive_clean = df_archive.copy()
          image_pred_clean= image_pred.copy()
          tweets_clean = tweets_df.copy()
```

1.9.1 Issue #1: change timestamp in df_archive_clean to datetime

Define: changing timestamp to datetime

Code

```
In [160]: df_archive_clean['timestamp'] = pd.to_datetime(df_archive_clean['timestamp'])
```

Test

```
In [161]: df_archive_clean.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2356 entries, 0 to 2355
Data columns (total 17 columns):
tweet_id                      2356 non-null int64
in_reply_to_status_id         78 non-null float64
in_reply_to_user_id           78 non-null float64
timestamp                     2356 non-null datetime64[ns]
source                        2356 non-null object
text                          2356 non-null object
retweeted_status_id           181 non-null float64
retweeted_status_user_id      181 non-null float64
retweeted_status_timestamp    181 non-null object
expanded_urls                 2297 non-null object
rating_numerator              2356 non-null int64
rating_denominator            2356 non-null int64
name                          2356 non-null object
doggo                         2356 non-null object
```

```
floofer                       2356 non-null object
pupper                        2356 non-null object
puppo                         2356 non-null object
dtypes: datetime64[ns](1), float64(4), int64(3), object(9)
memory usage: 313.0+ KB
```

1.9.2 Issue 2 and 3: droping null values

Define: dropping null values in, in_reply_to_status_id and in_reply_to_user_id

Code

```
In [162]: #this function gets the column index of non-null values of the dataframe passed into i
          def get_index(df_name, col):
              return df_name[df_name[col].notnull()].index
```

```
In [163]: # this function drops row or column that's passed into the function
          def drop(dframe, rw_cl, axis):
              dframe.drop(rw_cl, axis=axis, inplace=True)
```

```
In [164]: # getting the index of non-null values in df1_archive['in_reply_to_user_id']
          reply_id_index = get_index(df_archive_clean,'in_reply_to_user_id')
          reply_id_index
```

```
Out[164]: Int64Index([  30,   55,   64,  113,  148,  149,  179,  184,  186,  188,  189,
                       218,  228,  234,  251,  274,  290,  291,  313,  342,  346,  387,
                       409,  427,  498,  513,  565,  570,  576,  611,  701,  843,  857,
                       967, 1005, 1016, 1018, 1080, 1127, 1295, 1330, 1339, 1345, 1356,
                      1446, 1452, 1464, 1474, 1479, 1497, 1501, 1523, 1598, 1605, 1618,
                      1630, 1634, 1663, 1689, 1774, 1819, 1842, 1844, 1852, 1866, 1882,
                      1885, 1892, 1895, 1905, 1914, 1940, 2036, 2038, 2149, 2169, 2189,
                      2298],
                     dtype='int64')
```

```
In [165]: #dropping the rows of non-null value in in_reply_to_user_id
          drop(df_archive_clean, reply_id_index,0)
```

1.9.3 Test

```
In [166]: df_archive_clean.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 2278 entries, 0 to 2355
Data columns (total 17 columns):
tweet_id                      2278 non-null int64
in_reply_to_status_id         0 non-null float64
in_reply_to_user_id           0 non-null float64
timestamp                     2278 non-null datetime64[ns]
```

```
source                        2278 non-null object
text                          2278 non-null object
retweeted_status_id           181 non-null float64
retweeted_status_user_id      181 non-null float64
retweeted_status_timestamp    181 non-null object
expanded_urls                 2274 non-null object
rating_numerator              2278 non-null int64
rating_denominator            2278 non-null int64
name                          2278 non-null object
doggo                         2278 non-null object
floofer                       2278 non-null object
pupper                        2278 non-null object
puppo                         2278 non-null object
dtypes: datetime64[ns](1), float64(4), int64(3), object(9)
memory usage: 320.3+ KB
```

1.9.4 Issue 4 and 5. Dropping null values

Define : dropping null values in retweeted_status_id and retweeted_status_user_id and retweeted_status_timestamp

1.9.5 code

In [167]:
```
#getting the index of non-null values in df1_archive for retweeted_status_user_id
retweeted_status_id_index = get_index(df_archive_clean,'retweeted_status_user_id')
retweeted_status_id_index
```

Out[167]:
```
Int64Index([  19,   32,   36,   68,   73,   74,   78,   91,   95,   97,
            ...
             926,  937,  943,  949, 1012, 1023, 1043, 1242, 2259, 2260],
           dtype='int64', length=181)
```

In [168]:
```
# getting index of each variables passed into get_index
retweeted_user_id = get_index(df_archive_clean,'retweeted_status_user_id')
retweeted_timestamp = get_index(df_archive_clean, 'retweeted_status_timestamp')
```

In [169]:
```
# checking if the three variable has same index as rempving one affects all if true
a = retweeted_user_id == retweeted_timestamp & retweeted_status_id_index
for t in a:
    if t == False:
      print(t)
else:
    print('all true')
```

```
all true
```

In [170]:
```
# dropping the rows of non-null value in retweeted_status_user_id
drop(df_archive_clean, retweeted_status_id_index,axis=0)
```

1.9.6 Test

```
In [171]: df_archive_clean.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 2097 entries, 0 to 2355
Data columns (total 17 columns):
tweet_id                      2097 non-null int64
in_reply_to_status_id         0 non-null float64
in_reply_to_user_id           0 non-null float64
timestamp                     2097 non-null datetime64[ns]
source                        2097 non-null object
text                          2097 non-null object
retweeted_status_id           0 non-null float64
retweeted_status_user_id      0 non-null float64
retweeted_status_timestamp    0 non-null object
expanded_urls                 2094 non-null object
rating_numerator              2097 non-null int64
rating_denominator            2097 non-null int64
name                          2097 non-null object
doggo                         2097 non-null object
floofer                       2097 non-null object
pupper                        2097 non-null object
puppo                         2097 non-null object
dtypes: datetime64[ns](1), float64(4), int64(3), object(9)
memory usage: 294.9+ KB
```

1.9.7 Issue 6: Dropping irrelevant columns

1.9.8 code

```
In [172]: #creating list of columns to be dropped
          columns = ['in_reply_to_status_id', 'in_reply_to_user_id', 'retweeted_status_id', 'ret
```

```
In [173]: # dropping columns
          drop(df_archive_clean, columns, axis=1)
```

Test

```
In [174]: df_archive_clean.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 2097 entries, 0 to 2355
Data columns (total 12 columns):
tweet_id              2097 non-null int64
timestamp             2097 non-null datetime64[ns]
source                2097 non-null object
text                  2097 non-null object
expanded_urls         2094 non-null object
```

```
rating_numerator      2097 non-null int64
rating_denominator    2097 non-null int64
name                  2097 non-null object
doggo                 2097 non-null object
floofer               2097 non-null object
pupper                2097 non-null object
puppo                 2097 non-null object
dtypes: datetime64[ns](1), int64(3), object(8)
memory usage: 213.0+ KB
```

1.9.9 Issue 7: source contains anchor tags, re-extracting sourecs

1.9.10 Define: re-extracting source

code

```
In [175]: # increasing length information the display in the dataframe so regex can used to extr
          df_archive_clean.style.set_properties(subset=['timestamp'], **{'width': '300px'})

Out[175]: <pandas.io.formats.style.Styler at 0x7f82b9c84860>

In [176]: df_archive_clean.head()

Out[176]:              tweet_id           timestamp  \
          0  892420643555336193 2017-08-01 16:23:56
          1  892177421306343426 2017-08-01 00:17:27
          2  891815181378084864 2017-07-31 00:18:03
          3  891689557279858688 2017-07-30 15:58:51
          4  891327558926688256 2017-07-29 16:00:24

                                                        source  \
          0  <a href="http://twitter.com/download/iphone" r...
          1  <a href="http://twitter.com/download/iphone" r...
          2  <a href="http://twitter.com/download/iphone" r...
          3  <a href="http://twitter.com/download/iphone" r...
          4  <a href="http://twitter.com/download/iphone" r...

                                                          text  \
          0  This is Phineas. He's a mystical boy. Only eve...
          1  This is Tilly. She's just checking pup on you...
          2  This is Archie. He is a rare Norwegian Pouncin...
          3  This is Darla. She commenced a snooze mid meal...
          4  This is Franklin. He would like you to stop ca...

                                                 expanded_urls  rating_numerator  \
          0  https://twitter.com/dog_rates/status/892420643...                13
          1  https://twitter.com/dog_rates/status/892177421...                13
          2  https://twitter.com/dog_rates/status/891815181...                12
```

```
3  https://twitter.com/dog_rates/status/891689557...                 13
4  https://twitter.com/dog_rates/status/891327558...                 12

   rating_denominator      name doggo floofer pupper puppo
0                  10   Phineas  None    None   None  None
1                  10     Tilly  None    None   None  None
2                  10    Archie  None    None   None  None
3                  10     Darla  None    None   None  None
4                  10  Franklin  None    None   None  None
```

In [177]:
```
# df_archive_clean.source.value_counts()
```

In [178]:
```
#df_archive_clean.source[1]
```

In [179]:
```
# extraction using regex expression
df_archive_clean['source'] =  df_archive_clean['source'].str.extract('^<a.+>(.+)</a>$'
```

Test

In [180]:
```
df_archive_clean.head()
```

Out[180]:
```
             tweet_id           timestamp              source  \
0  892420643555336193 2017-08-01 16:23:56  Twitter for iPhone
1  892177421306343426 2017-08-01 00:17:27  Twitter for iPhone
2  891815181378084864 2017-07-31 00:18:03  Twitter for iPhone
3  891689557279858688 2017-07-30 15:58:51  Twitter for iPhone
4  891327558926688256 2017-07-29 16:00:24  Twitter for iPhone

                                                text  \
0  This is Phineas. He's a mystical boy. Only eve...
1  This is Tilly. She's just checking pup on you...
2  This is Archie. He is a rare Norwegian Pouncin...
3  This is Darla. She commenced a snooze mid meal...
4  This is Franklin. He would like you to stop ca...

                                       expanded_urls  rating_numerator  \
0  https://twitter.com/dog_rates/status/892420643...                 13
1  https://twitter.com/dog_rates/status/892177421...                 13
2  https://twitter.com/dog_rates/status/891815181...                 12
3  https://twitter.com/dog_rates/status/891689557...                 13
4  https://twitter.com/dog_rates/status/891327558...                 12

   rating_denominator      name doggo floofer pupper puppo
0                  10   Phineas  None    None   None  None
1                  10     Tilly  None    None   None  None
2                  10    Archie  None    None   None  None
3                  10     Darla  None    None   None  None
4                  10  Franklin  None    None   None  None
```

1.9.11 Issue 8: Extracting properly numerator and denominator in ratings

1.9.12 code

```
In [181]: df_archive_clean['rating_numerator'] = df_archive_clean['text'].str.extract('(\d+\.*\d

In [182]: df_archive_clean['rating_denominator'] = df_archive_clean['text'].str.extract('(\d+\.*
```

1.9.13 Test

```
In [183]: df_archive_clean['rating_numerator'].sample(10),df_archive_clean['rating_denominator']

Out[183]: (2176     9
           1370    12
           1040    12
           1742    11
           752     11
           1707    10
           2254     9
           1462    10
           1808     5
           1919     8
           Name: rating_numerator, dtype: object, 1051    10
           918     10
           1337    10
           1588    10
           1290    10
           1553    10
           1359    10
           483     10
           1824    10
           248     10
           Name: rating_denominator, dtype: object)
```

1.9.14 Issue 9: making dog stages into one column

```
In [184]: df_archive_clean.sample(2)

Out[184]:                 tweet_id           timestamp              source  \
          2267  667524857454854144 2015-11-20 02:08:22  Twitter Web Client
          52    882045870035918850 2017-07-04 01:18:17  Twitter for iPhone

                                                             text  \
          2267  Another topnotch dog. His name is Big Jumpy Ra...
          52    This is Koko. Her owner, inspired by Barney, r...

                                                    expanded_urls rating_numerator  \
          2267  https://twitter.com/dog_rates/status/667524857...               12
          52    https://twitter.com/dog_rates/status/882045870...               13
```

```
     rating_denominator  name doggo floofer pupper puppo
2267                 10  None  None    None   None  None
52                   10  Koko  None    None   None  None
```

In [185]:
```
def join_df(x):
    return ', '.join(x.dropna().astype(str))
```

In [186]:
```
#creating list of dog stages
dog_stages = ['doggo', 'floofer', 'pupper', 'puppo']

# replace none with nan in dog stages
df_archive_clean[dog_stages] = df_archive_clean[dog_stages].replace('None', np.nan)
```

In [187]:
```
df_archive_clean.head(3)
```

Out[187]:
```
             tweet_id           timestamp              source  \
0  892420643555336193 2017-08-01 16:23:56  Twitter for iPhone
1  892177421306343426 2017-08-01 00:17:27  Twitter for iPhone
2  891815181378084864 2017-07-31 00:18:03  Twitter for iPhone

                                                text  \
0  This is Phineas. He's a mystical boy. Only eve...
1  This is Tilly. She's just checking pup on you...
2  This is Archie. He is a rare Norwegian Pouncin...

                                       expanded_urls rating_numerator  \
0  https://twitter.com/dog_rates/status/892420643...               13
1  https://twitter.com/dog_rates/status/892177421...               13
2  https://twitter.com/dog_rates/status/891815181...               12

  rating_denominator     name doggo floofer pupper puppo
0                 10  Phineas   NaN     NaN    NaN   NaN
1                 10    Tilly   NaN     NaN    NaN   NaN
2                 10   Archie   NaN     NaN    NaN   NaN
```

In [188]:
```
df_archive_clean['dog_stages'] = df_archive_clean[dog_stages].apply(join_df, axis=1)

#replace empty string with nan
df_archive_clean['dog_stages'] = df_archive_clean['dog_stages'].replace('', np.nan)

#drop the four column: doggo, floofer, pupper, puppo
drop(df_archive_clean, dog_stages, axis=1)
```

test

In [189]:
```
df_archive_clean.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 2097 entries, 0 to 2355
```

```
Data columns (total 9 columns):
tweet_id              2097 non-null int64
timestamp             2097 non-null datetime64[ns]
source                2097 non-null object
text                  2097 non-null object
expanded_urls         2094 non-null object
rating_numerator      2097 non-null object
rating_denominator    2097 non-null object
name                  2097 non-null object
dog_stages            336 non-null object
dtypes: datetime64[ns](1), int64(1), object(7)
memory usage: 243.8+ KB
```

In [190]:
```
df_archive_clean['dog_stages'].value_counts()
```

Out[190]:
```
pupper            221
doggo              72
puppo              23
floofer             9
doggo, pupper       9
doggo, floofer      1
doggo, puppo        1
Name: dog_stages, dtype: int64
```

In [191]:
```
df_archive_clean.head(2)
```

Out[191]:
```
             tweet_id           timestamp              source  \
0  892420643555336193 2017-08-01 16:23:56  Twitter for iPhone
1  892177421306343426 2017-08-01 00:17:27  Twitter for iPhone

                                                text  \
0  This is Phineas. He's a mystical boy. Only eve...
1  This is Tilly. She's just checking pup on you...

                                       expanded_urls rating_numerator  \
0  https://twitter.com/dog_rates/status/892420643...               13
1  https://twitter.com/dog_rates/status/892177421...               13

  rating_denominator     name dog_stages
0                 10  Phineas        NaN
1                 10    Tilly        NaN
```

1.9.15 Issue 10: removing tweets that contains no dog.

- "we only rate dogs" is a phrase used for images or tweet not dog

1.9.16 code

In [192]:
```
df_archive_clean['text'].sample(30)
```

```
Out[192]: 832     Say hello to Oakley and Charlie. They're convi...
          1164    This is Jangle. She's addicted to broccoli. It...
          1333    This is Cooper. He only wakes up to switch gea...
          2242    Wow. Armored dog here. Ready for battle. Face ...
          1205    This is Bubbles. He's a Yorkshire Piccolope. 1...
          1502    This is Teddy. His head is too heavy. 13/10 (v...
          939     So this just changed my life. 13/10 please enj...
          464     Meet Strudel. He's rather h*ckin pupset that y...
          1181    This is Pippa. She managed to start the car bu...
          242     This is Vixen. He really likes bananas. Steals...
          1734    This pup's name is Sabertooth (parents must be...
          1029    This is Percy. He fell asleep at the wheel. Ir...
          1586    This pupper forgot how to walk. 12/10 happens ...
          2348    Here is a Siberian heavily armored polar bear ...
          2272    Two dogs in this one. Both are rare Jujitsu Py...
          932     This is Charlie. He pouts until he gets to go ...
          1182    This is Sadie. She is prepared for battle. 10/...
          2108    This dog can't see its haters. 11/10 https://t...
          2307    12/10 simply brilliant pup https://t.co/V6ZzG4...
          2117             Meet Herb. 12/10 https://t.co/tLRyYvCci3
          1662    This is Darrel. He just robbed a 7/11 and is i...
          1854    Seriously guys?! Only send in dogs. I only rat...
          1977    This is Schnozz. He's had a blurred tail since...
          1969    Take a moment and appreciate how these two dog...
          1259    We  only  rate  dogs. Pls stop sending i...
          2170    This is Sully. He's a Leviticus Galapagos. Ver...
          642     This is Maude. She's the h*ckin happiest wasp ...
          489     This is Chubbs. He dug a hole and now he's stu...
          1262    This is Tater. His underbite is fierce af. Doe...
          67      This is Jack AKA Stephen Furry. You're not sco...
          Name: text, dtype: object
```

```
In [193]: df_archive_clean.iloc[59]
```

```
Out[193]: tweet_id                                          879492040517615616
          timestamp                                        2017-06-27 00:10:17
          source                                            Twitter for iPhone
          text                   This is Bailey. He thinks you should measure e...
          expanded_urls          https://twitter.com/dog_rates/status/879492040...
          rating_numerator                                                  12
          rating_denominator                                                10
          name                                                          Bailey
          dog_stages                                                       NaN
          Name: 65, dtype: object
```

```
In [194]: df_archive_clean.iloc[103]
```

```
Out[194]: tweet_id                                          869702957897576449
          timestamp                                        2017-05-30 23:51:58
```

```
source                                              Twitter for iPhone
text                     Meet Stanley. He likes road trips. Will shift ...
expanded_urls            https://twitter.com/dog_rates/status/869702957...
rating_numerator                                                    13
rating_denominator                                                  10
name                                                           Stanley
dog_stages                                                         NaN
Name: 120, dtype: object
```

```
In [195]: # getting tweets not dogs
          not_dogs = df_archive_clean[df_archive_clean['text'].str.match('.*only rate dog')].ind
```

```
In [196]: # dropping rows of tweet not dog in tweet_archive
          drop(df_archive_clean, not_dogs, axis=0)
```

test

```
In [197]: #checking if there's still tweet or image not dog related
          df_archive_clean[df_archive_clean['text'].str.match('.*only rate dog')]
```

```
Out[197]: Empty DataFrame
          Columns: [tweet_id, timestamp, source, text, expanded_urls, rating_numerator, rating_d
          Index: []
```

1.9.17 Isuue 11: renaming column name id to tweet_id in tweets_

1.9.18 code

```
In [198]: tweets_clean.rename(columns={'id': 'tweet_id'}, inplace=True)
```

1.9.19 Test

```
In [199]: tweets_clean.head()
```

Out[199]:

	tweet_id	retweet_count	favorite_count
0	892420643555336193	8853	39467
1	892177421306343426	6514	33819
2	891815181378084864	4328	25461
3	891689557279858688	8964	42908
4	891327558926688256	9774	41048

1.9.20 Issue 12: selecting images that are only dogs

code

```
In [200]: image_pred_clean.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2075 entries, 0 to 2074
Data columns (total 12 columns):
```

```
tweet_id    2075 non-null int64
jpg_url     2075 non-null object
img_num     2075 non-null int64
p1          2075 non-null object
p1_conf     2075 non-null float64
p1_dog      2075 non-null bool
p2          2075 non-null object
p2_conf     2075 non-null float64
p2_dog      2075 non-null bool
p3          2075 non-null object
p3_conf     2075 non-null float64
p3_dog      2075 non-null bool
dtypes: bool(3), float64(3), int64(2), object(4)
memory usage: 152.1+ KB
```

In [201]:
```
# getting instances where
c1 =image_pred_clean['p1_dog'] == True
```

In [202]:
```
c2 =image_pred_clean['p2_dog'] == True
```

In [203]:
```
c3 =image_pred_clean['p3_dog'] == True
```

In [204]:
```
# defining condition for selection
condition = [c1, c2, c3]
```

In [205]:
```
# set the choice list based on the selection conditions for predicted breed
breed = [image_pred_clean['p1'], image_pred_clean['p2'],image_pred_clean['p3']]
```

In [206]:
```
#set the choice order for confidence level based on the selection condition
confidence = [image_pred_clean['p1_conf'], image_pred_clean['p2_conf'], image_pred_cle
```

In [207]:
```
# select the predicted breed based on condition set initially, creating and reading it
image_pred_clean['breed'] = np.select(condition, breed, default = 'none')
```

In [208]:
```
# select the predicted confidence based on condition set initially, creating and readi
image_pred_clean['confidence'] = np.select(condition, confidence, default=0)
```

1.9.21 Issue 13: Dropping irrelevant column in image prediction

1.9.22 code

In [209]:
```
image_pred_clean.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2075 entries, 0 to 2074
Data columns (total 14 columns):
tweet_id      2075 non-null int64
jpg_url       2075 non-null object
img_num       2075 non-null int64
```

```
p1            2075 non-null object
p1_conf       2075 non-null float64
p1_dog        2075 non-null bool
p2            2075 non-null object
p2_conf       2075 non-null float64
p2_dog        2075 non-null bool
p3            2075 non-null object
p3_conf       2075 non-null float64
p3_dog        2075 non-null bool
breed         2075 non-null object
confidence    2075 non-null float64
dtypes: bool(3), float64(4), int64(2), object(5)
memory usage: 184.5+ KB
```

```
In [210]: # defining columns to be drooped
          columns= ['p1','p1_conf', 'p1_dog', 'p2','p2_conf', 'p2_dog', 'p3','p3_conf', 'p3_dog'
```

```
In [211]: # dropping the columns
          drop(image_pred_clean, columns, axis=1)
```

1.9.23 test

```
In [212]: image_pred_clean.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2075 entries, 0 to 2074
Data columns (total 4 columns):
tweet_id      2075 non-null int64
jpg_url       2075 non-null object
breed         2075 non-null object
confidence    2075 non-null float64
dtypes: float64(1), int64(1), object(2)
memory usage: 64.9+ KB
```

1.9.24 Issue 14: changing rating_numerator and denominator to float

```
In [213]: # converting rating_numerator and rating_denominator
          df_archive_clean['rating_numerator'] = pd.to_numeric(df_archive_clean['rating_numerato
          df_archive_clean['rating_denominator'] = pd.to_numeric(df_archive_clean['rating_denomi
```

1.9.25 Test

```
In [214]: df_archive_clean.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 2043 entries, 0 to 2355
Data columns (total 9 columns):
```

```
tweet_id               2043 non-null int64
timestamp              2043 non-null datetime64[ns]
source                 2043 non-null object
text                   2043 non-null object
expanded_urls          2040 non-null object
rating_numerator       2043 non-null float64
rating_denominator     2043 non-null int64
name                   2043 non-null object
dog_stages             336 non-null object
dtypes: datetime64[ns](1), float64(1), int64(2), object(5)
memory usage: 159.6+ KB
```

1.9.26 Issue 15: merging dataframe into one dataframe

1.9.27 code

```
In [215]: df_archive_clean.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 2043 entries, 0 to 2355
Data columns (total 9 columns):
tweet_id               2043 non-null int64
timestamp              2043 non-null datetime64[ns]
source                 2043 non-null object
text                   2043 non-null object
expanded_urls          2040 non-null object
rating_numerator       2043 non-null float64
rating_denominator     2043 non-null int64
name                   2043 non-null object
dog_stages             336 non-null object
dtypes: datetime64[ns](1), float64(1), int64(2), object(5)
memory usage: 159.6+ KB
```

```
In [216]: combine_df = pd.merge(df_archive_clean, tweets_clean, on='tweet_id', how = 'left')
```

```
In [217]: df = pd.merge(combine_df, image_pred_clean, on='tweet_id', how = 'inner')
```

```
In [218]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 1917 entries, 0 to 1916
Data columns (total 14 columns):
tweet_id               1917 non-null int64
timestamp              1917 non-null datetime64[ns]
source                 1917 non-null object
text                   1917 non-null object
expanded_urls          1917 non-null object
```

```
rating_numerator      1917 non-null float64
rating_denominator    1917 non-null int64
name                  1917 non-null object
dog_stages            303 non-null object
retweet_count         1917 non-null int64
favorite_count        1917 non-null int64
jpg_url               1917 non-null object
breed                 1917 non-null object
confidence            1917 non-null float64
dtypes: datetime64[ns](1), float64(2), int64(4), object(7)
memory usage: 224.6+ KB
```

saving master dataframe as twitter_archive_master.csv

code

```
In [219]: # saving dataframe
          df.to_csv('twitter_archive_master.csv', index=False)
```

test

```
In [220]: pd.read_csv('twitter_archive_master.csv')
```

```
Out[220]:                tweet_id            timestamp              source  \
          0     892420643555336193  2017-08-01 16:23:56  Twitter for iPhone
          1     892177421306343426  2017-08-01 00:17:27  Twitter for iPhone
          2     891815181378084864  2017-07-31 00:18:03  Twitter for iPhone
          3     891689557279858688  2017-07-30 15:58:51  Twitter for iPhone
          4     891327558926688256  2017-07-29 16:00:24  Twitter for iPhone
          5     891087950875897856  2017-07-29 00:08:17  Twitter for iPhone
          6     890971913173991426  2017-07-28 16:27:12  Twitter for iPhone
          7     890729181411237888  2017-07-28 00:22:40  Twitter for iPhone
          8     890609185150312448  2017-07-27 16:25:51  Twitter for iPhone
          9     890240255349198849  2017-07-26 15:59:51  Twitter for iPhone
          10    890006608113172480  2017-07-26 00:31:25  Twitter for iPhone
          11    889880896479866881  2017-07-25 16:11:53  Twitter for iPhone
          12    889665388333682689  2017-07-25 01:55:32  Twitter for iPhone
          13    889638837579907072  2017-07-25 00:10:02  Twitter for iPhone
          14    889531135344209921  2017-07-24 17:02:04  Twitter for iPhone
          15    889278841981685760  2017-07-24 00:19:32  Twitter for iPhone
          16    888917238123831296  2017-07-23 00:22:39  Twitter for iPhone
          17    888804989199671297  2017-07-22 16:56:37  Twitter for iPhone
          18    888554962724278272  2017-07-22 00:23:06  Twitter for iPhone
          19    888078434458587136  2017-07-20 16:49:33  Twitter for iPhone
          20    887705289381826560  2017-07-19 16:06:48  Twitter for iPhone
          21    887517139158093824  2017-07-19 03:39:09  Twitter for iPhone
          22    887473957103951883  2017-07-19 00:47:34  Twitter for iPhone
```

```
23    887343217045368832  2017-07-18 16:08:03  Twitter for iPhone
24    886983233522544640  2017-07-17 16:17:36  Twitter for iPhone
25    886736880519319552  2017-07-16 23:58:41  Twitter for iPhone
26    886680336477933568  2017-07-16 20:14:00  Twitter for iPhone
27    886366144734445568  2017-07-15 23:25:31  Twitter for iPhone
28    886258384151887873  2017-07-15 16:17:19  Twitter for iPhone
29    885984800019947520  2017-07-14 22:10:11  Twitter for iPhone
...                  ...                  ...                 ...
1887  666411507551481857  2015-11-17 00:24:19  Twitter for iPhone
1888  666407126856765440  2015-11-17 00:06:54  Twitter for iPhone
1889  666396247373291520  2015-11-16 23:23:41  Twitter for iPhone
1890  666373753744588802  2015-11-16 21:54:18  Twitter for iPhone
1891  666362758909284353  2015-11-16 21:10:36  Twitter for iPhone
1892  666353288456101888  2015-11-16 20:32:58  Twitter for iPhone
1893  666345417576210432  2015-11-16 20:01:42  Twitter for iPhone
1894  666337882303524864  2015-11-16 19:31:45  Twitter for iPhone
1895  666293911632134144  2015-11-16 16:37:02  Twitter for iPhone
1896  666287406224695296  2015-11-16 16:11:11  Twitter for iPhone
1897  666273097616637952  2015-11-16 15:14:19  Twitter for iPhone
1898  666268910803644416  2015-11-16 14:57:41  Twitter for iPhone
1899  666104133288665088  2015-11-16 04:02:55  Twitter for iPhone
1900  666102155909144576  2015-11-16 03:55:04  Twitter for iPhone
1901  666099513787052032  2015-11-16 03:44:34  Twitter for iPhone
1902  666094000022159362  2015-11-16 03:22:39  Twitter for iPhone
1903  666082916733198337  2015-11-16 02:38:37  Twitter for iPhone
1904  666073100786774016  2015-11-16 01:59:36  Twitter for iPhone
1905  666071193221509120  2015-11-16 01:52:02  Twitter for iPhone
1906  666063827256086533  2015-11-16 01:22:45  Twitter for iPhone
1907  666058600524156928  2015-11-16 01:01:59  Twitter for iPhone
1908  666057090499244032  2015-11-16 00:55:59  Twitter for iPhone
1909  666055525042405380  2015-11-16 00:49:46  Twitter for iPhone
1910  666051853826850816  2015-11-16 00:35:11  Twitter for iPhone
1911  666050758794694657  2015-11-16 00:30:50  Twitter for iPhone
1912  666049248165822465  2015-11-16 00:24:50  Twitter for iPhone
1913  666044226329800704  2015-11-16 00:04:52  Twitter for iPhone
1914  666033412701032449  2015-11-15 23:21:54  Twitter for iPhone
1915  666029285002620928  2015-11-15 23:05:30  Twitter for iPhone
1916  666020888022790149  2015-11-15 22:32:08  Twitter for iPhone

                                                   text  \
0     This is Phineas. He's a mystical boy. Only eve...
1     This is Tilly. She's just checking pup on you...
2     This is Archie. He is a rare Norwegian Pouncin...
3     This is Darla. She commenced a snooze mid meal...
4     This is Franklin. He would like you to stop ca...
5     Here we have a majestic great white breaching ...
6     Meet Jax. He enjoys ice cream so much he gets ...
7     When you watch your owner call another dog a g...
```

```
8       This is Zoey. She doesn't want to be one of th...
9       This is Cassie. She is a college pup. Studying...
10      This is Koda. He is a South Australian decksha...
11      This is Bruno. He is a service shark. Only get...
12      Here's a puppo that seems to be on the fence a...
13      This is Ted. He does his best. Sometimes that'...
14      This is Stuart. He's sporting his favorite fan...
15      This is Oliver. You're witnessing one of his m...
16      This is Jim. He found a fren. Taught him how t...
17      This is Zeke. He has a new stick. Very proud o...
18      This is Ralphus. He's powering up. Attempting ...
19      This is Gerald. He was just told he didn't get...
20      This is Jeffrey. He has a monopoly on the pool...
21      I've yet to rate a Venezuelan Hover Wiener. Th...
22      This is Canela. She attempted some fancy porch...
23      You may not have known you needed to see this ...
24      This is Maya. She's very shy. Rarely leaves he...
25      This is Mingus. He's a wonderful father to his...
26      This is Derek. He's late for a dog meeting. 13...
27      This is Roscoe. Another pupper fallen victim t...
28      This is Waffles. His doggles are pupside down...
29      Viewer discretion advised. This is Jimbo. He w...
...                                                   ...
1887    This is quite the dog. Gets really excited whe...
1888    This is a southern Vesuvius bumblegruff. Can d...
1889    Oh goodness. A super rare northeast Qdoba kang...
1890    Those are sunglasses and a jean jacket. 11/10 ...
1891    Unique dog here. Very small. Lives in containe...
1892    Here we have a mixed Asiago from the Galápagos...
1893    Look at this jokester thinking seat belt laws ...
1894    This is an extremely rare horned Parthenon. No...
1895    This is a funny dog. Weird toes. Won't come do...
1896    This is an Albanian 3 1/2 legged  Episcopalian...
1897       Can take selfies 11/10 https://t.co/ws2AMaNwPW
1898    Very concerned about fellow dog trapped in com...
1899    Not familiar with this breed. No tail (weird)...
1900    Oh my. Here you are seeing an Adobe Setter giv...
1901    Can stand on stump for what seems like a while...
1902    This appears to be a Mongolian Presbyterian mi...
1903    Here we have a well-established sunblockerspan...
1904    Let's hope this flight isn't Malaysian (lol). ...
1905    Here we have a northern speckled Rhododendron...
1906    This is the happiest dog you will ever see. Ve...
1907    Here is the Rand Paul of retrievers folks! He'...
1908    My oh my. This is a rare blond Canadian terrie...
1909    Here is a Siberian heavily armored polar bear ...
1910    This is an odd dog. Hard on the outside but lo...
1911    This is a truly beautiful English Wilson Staff...
```

```
1912  Here we have a 1949 1st generation vulpix. Enj...
1913  This is a purebred Piers Morgan. Loves to Netf...
1914  Here is a very happy pup. Big fan of well-main...
1915  This is a western brown Mitsubishi terrier. Up...
1916  Here we have a Japanese Irish Setter. Lost eye...

                                         expanded_urls  rating_numerator  \
0     https://twitter.com/dog_rates/status/892420643...              13.0
1     https://twitter.com/dog_rates/status/892177421...              13.0
2     https://twitter.com/dog_rates/status/891815181...              12.0
3     https://twitter.com/dog_rates/status/891689557...              13.0
4     https://twitter.com/dog_rates/status/891327558...              12.0
5     https://twitter.com/dog_rates/status/891087950...              13.0
6     https://gofundme.com/ydvmve-surgery-for-jax,ht...              13.0
7     https://twitter.com/dog_rates/status/890729181...              13.0
8     https://twitter.com/dog_rates/status/890609185...              13.0
9     https://twitter.com/dog_rates/status/890240255...              14.0
10    https://twitter.com/dog_rates/status/890006608...              13.0
11    https://twitter.com/dog_rates/status/889880896...              13.0
12    https://twitter.com/dog_rates/status/889665388...              13.0
13    https://twitter.com/dog_rates/status/889638837...              12.0
14    https://twitter.com/dog_rates/status/889531135...              13.0
15    https://twitter.com/dog_rates/status/889278841...              13.0
16    https://twitter.com/dog_rates/status/888917238...              12.0
17    https://twitter.com/dog_rates/status/888804989...              13.0
18    https://twitter.com/dog_rates/status/888554962...              13.0
19    https://twitter.com/dog_rates/status/888078434...              12.0
20    https://twitter.com/dog_rates/status/887705289...              13.0
21    https://twitter.com/dog_rates/status/887517139...              14.0
22    https://twitter.com/dog_rates/status/887473957...              13.0
23    https://twitter.com/dog_rates/status/887343217...              13.0
24    https://twitter.com/dog_rates/status/886983233...              13.0
25    https://www.gofundme.com/mingusneedsus,https:/...              13.0
26    https://twitter.com/dog_rates/status/886680336...              13.0
27    https://twitter.com/dog_rates/status/886366144...              12.0
28    https://twitter.com/dog_rates/status/886258384...              13.0
29    https://twitter.com/dog_rates/status/885984800...              12.0
...                                                 ...               ...
1887  https://twitter.com/dog_rates/status/666411507...               2.0
1888  https://twitter.com/dog_rates/status/666407126...               7.0
1889  https://twitter.com/dog_rates/status/666396247...               9.0
1890  https://twitter.com/dog_rates/status/666373753...              11.0
1891  https://twitter.com/dog_rates/status/666362758...               6.0
1892  https://twitter.com/dog_rates/status/666353288...               8.0
1893  https://twitter.com/dog_rates/status/666345417...              10.0
1894  https://twitter.com/dog_rates/status/666337882...               9.0
1895  https://twitter.com/dog_rates/status/666293911...               3.0
1896  https://twitter.com/dog_rates/status/666287406...               1.0
```

```
1897  https://twitter.com/dog_rates/status/666273097...            11.0
1898  https://twitter.com/dog_rates/status/666268910...            10.0
1899  https://twitter.com/dog_rates/status/666104133...             1.0
1900  https://twitter.com/dog_rates/status/666102155...            11.0
1901  https://twitter.com/dog_rates/status/666099513...             8.0
1902  https://twitter.com/dog_rates/status/666094000...             9.0
1903  https://twitter.com/dog_rates/status/666082916...             6.0
1904  https://twitter.com/dog_rates/status/666073100...            10.0
1905  https://twitter.com/dog_rates/status/666071193...             9.0
1906  https://twitter.com/dog_rates/status/666063827...            10.0
1907  https://twitter.com/dog_rates/status/666058600...             8.0
1908  https://twitter.com/dog_rates/status/666057090...             9.0
1909  https://twitter.com/dog_rates/status/666055525...            10.0
1910  https://twitter.com/dog_rates/status/666051853...             2.0
1911  https://twitter.com/dog_rates/status/666050758...            10.0
1912  https://twitter.com/dog_rates/status/666049248...             5.0
1913  https://twitter.com/dog_rates/status/666044226...             6.0
1914  https://twitter.com/dog_rates/status/666033412...             9.0
1915  https://twitter.com/dog_rates/status/666029285...             7.0
1916  https://twitter.com/dog_rates/status/666020888...             8.0

      rating_denominator      name dog_stages  retweet_count  favorite_count  \
0                     10   Phineas        NaN           8853           39467
1                     10     Tilly        NaN           6514           33819
2                     10    Archie        NaN           4328           25461
3                     10     Darla        NaN           8964           42908
4                     10  Franklin        NaN           9774           41048
5                     10      None        NaN           3261           20562
6                     10       Jax        NaN           2158           12041
7                     10      None        NaN          16716           56848
8                     10      Zoey        NaN           4429           28226
9                     10    Cassie      doggo           7711           32467
10                    10      Koda        NaN           7624           31166
11                    10     Bruno        NaN           5156           28268
12                    10      None      puppo           8538           38818
13                    10       Ted        NaN           4735           27672
14                    10    Stuart      puppo           2321           15359
15                    10    Oliver        NaN           5637           25652
16                    10       Jim        NaN           4709           29611
17                    10      Zeke        NaN           4559           26080
18                    10   Ralphus        NaN           3732           20290
19                    10    Gerald        NaN           3653           22201
20                    10   Jeffrey        NaN           5609           30779
21                    10      such        NaN          12082           46959
22                    10    Canela        NaN          18781           69871
23                    10      None        NaN          10737           34222
24                    10      Maya        NaN           8084           35859
25                    10    Mingus        NaN           3443           12306
```

```
26                      10     Derek      NaN          4610          22798
27                      10    Roscoe   pupper          3316          21524
28                      10   Waffles      NaN          6523          28469
29                      10     Jimbo      NaN          7097          33382
...                    ...       ...      ...           ...            ...
1887                    10     quite      NaN           339            459
1888                    10         a      NaN            44            113
1889                    10      None      NaN            92            172
1890                    10      None      NaN           100            194
1891                    10      None      NaN           595            804
1892                    10      None      NaN            77            229
1893                    10      None      NaN           146            307
1894                    10        an      NaN            96            204
1895                    10         a      NaN           368            522
1896                     2        an      NaN            71            152
1897                    10      None      NaN            82            184
1898                    10      None      NaN            37            108
1899                    10      None      NaN          6871          14765
1900                    10      None      NaN            16             81
1901                    10      None      NaN            73            164
1902                    10      None      NaN            79            169
1903                    10      None      NaN            47            121
1904                    10      None      NaN           174            335
1905                    10      None      NaN            67            154
1906                    10       the      NaN           232            496
1907                    10       the      NaN            61            115
1908                    10         a      NaN           146            304
1909                    10         a      NaN           261            448
1910                    10        an      NaN           879           1253
1911                    10         a      NaN            60            136
1912                    10      None      NaN            41            111
1913                    10         a      NaN           147            311
1914                    10         a      NaN            47            128
1915                    10         a      NaN            48            132
1916                    10      None      NaN           532           2535

                                                jpg_url  \
0        https://pbs.twimg.com/media/DGKD1-bXoAAIAUK.jpg
1        https://pbs.twimg.com/media/DGGmoV4XsAAUL6n.jpg
2        https://pbs.twimg.com/media/DGBdLU1WsAANxJ9.jpg
3        https://pbs.twimg.com/media/DF_q7IAWsAEuuN8.jpg
4        https://pbs.twimg.com/media/DF6hr6BUMAAzZgT.jpg
5        https://pbs.twimg.com/media/DF3HwyEWsAABqE6.jpg
6        https://pbs.twimg.com/media/DF1eOmZXUAALUcq.jpg
7        https://pbs.twimg.com/media/DFyBahAVwAAhUTd.jpg
8        https://pbs.twimg.com/media/DFwUU__XcAEpyXI.jpg
9        https://pbs.twimg.com/media/DFrEyVuW0AAO3t9.jpg
10       https://pbs.twimg.com/media/DFnwSY4WAAAMliS.jpg
```

11	https://pbs.twimg.com/media/DFl99B1WsAITKsg.jpg
12	https://pbs.twimg.com/media/DFi579UWsAAatzw.jpg
13	https://pbs.twimg.com/media/DFihzFfXsAYGDPR.jpg
14	https://pbs.twimg.com/media/DFg_2PVW0AEHN3p.jpg
15	https://pbs.twimg.com/ext_tw_video_thumb/88927...
16	https://pbs.twimg.com/media/DFYRgsOUQAARGhO.jpg
17	https://pbs.twimg.com/media/DFWra-3VYAA2piG.jpg
18	https://pbs.twimg.com/media/DFTH_O-UQAACu2O.jpg
19	https://pbs.twimg.com/media/DFMWn56WsAAkA7B.jpg
20	https://pbs.twimg.com/media/DFHDQBbXgAEqY7t.jpg
21	https://pbs.twimg.com/ext_tw_video_thumb/88751...
22	https://pbs.twimg.com/media/DFDw2tyUQAAAFke.jpg
23	https://pbs.twimg.com/ext_tw_video_thumb/88734...
24	https://pbs.twimg.com/media/DE8yicJW0AAAvBJ.jpg
25	https://pbs.twimg.com/media/DE5Se8FXcAAJFx4.jpg
26	https://pbs.twimg.com/media/DE4fEDzWAAAyHMM.jpg
27	https://pbs.twimg.com/media/DE0BTnQUwAApKEH.jpg
28	https://pbs.twimg.com/media/DEyfTG4UMAE4aE9.jpg
29	https://pbs.twimg.com/media/DEumeWWV0AA-Z61.jpg
...	...
1887	https://pbs.twimg.com/media/CT-RugiWIAELEaq.jpg
1888	https://pbs.twimg.com/media/CT-NvwmW4AAugGZ.jpg
1889	https://pbs.twimg.com/media/CT-D2ZHWIAA3gK1.jpg
1890	https://pbs.twimg.com/media/CT9vZEYWUAAlZ05.jpg
1891	https://pbs.twimg.com/media/CT9lXGsUcAAyUFt.jpg
1892	https://pbs.twimg.com/media/CT9cx0tUEAAhNN_.jpg
1893	https://pbs.twimg.com/media/CT9Vn7PWoAA_ZCM.jpg
1894	https://pbs.twimg.com/media/CT9OwFIWEAMuRje.jpg
1895	https://pbs.twimg.com/media/CT8mx7KW4AEQu8N.jpg
1896	https://pbs.twimg.com/media/CT8g3BpUEAAuFjg.jpg
1897	https://pbs.twimg.com/media/CT8T1mtUwAA3aqm.jpg
1898	https://pbs.twimg.com/media/CT8QCd1WEAADXws.jpg
1899	https://pbs.twimg.com/media/CT56LSZWoAAlJj2.jpg
1900	https://pbs.twimg.com/media/CT54YGiWUAEZnoK.jpg
1901	https://pbs.twimg.com/media/CT51-JJUEAA6hV8.jpg
1902	https://pbs.twimg.com/media/CT5w9gUW4AAsBNN.jpg
1903	https://pbs.twimg.com/media/CT5m4VGWEAAtKc8.jpg
1904	https://pbs.twimg.com/media/CT5d9DZXAAALcwe.jpg
1905	https://pbs.twimg.com/media/CT5cN_3WEAAlOoZ.jpg
1906	https://pbs.twimg.com/media/CT5Vg_wXIAAXfnj.jpg
1907	https://pbs.twimg.com/media/CT5Qw94XAAA_2dP.jpg
1908	https://pbs.twimg.com/media/CT5PY9OWoAAQGLo.jpg
1909	https://pbs.twimg.com/media/CT5N9tpXIAAifs1.jpg
1910	https://pbs.twimg.com/media/CT5KoJ1WoAAJash.jpg
1911	https://pbs.twimg.com/media/CT5Jof1WUAEuVxN.jpg
1912	https://pbs.twimg.com/media/CT5IQmsXIAAKY4A.jpg
1913	https://pbs.twimg.com/media/CT5Dr8HUEAA-lEu.jpg
1914	https://pbs.twimg.com/media/CT4521TWwAEvMyu.jpg

```
1915    https://pbs.twimg.com/media/CT42GRgUYAA5iDo.jpg
1916    https://pbs.twimg.com/media/CT4udn0WwAA0aMy.jpg

                            breed  confidence
0                            none    0.000000
1                       Chihuahua    0.323581
2                       Chihuahua    0.716012
3              Labrador_retriever    0.168086
4                          basset    0.555712
5        Chesapeake_Bay_retriever    0.425595
6                     Appenzeller    0.341703
7                      Pomeranian    0.566142
8                   Irish_terrier    0.487574
9                        Pembroke    0.511319
10                        Samoyed    0.957979
11                 French_bulldog    0.377417
12                       Pembroke    0.966327
13                 French_bulldog    0.991650
14               golden_retriever    0.953442
15                        whippet    0.626152
16               golden_retriever    0.714719
17               golden_retriever    0.469760
18                 Siberian_husky    0.700377
19                 French_bulldog    0.995026
20                         basset    0.821664
21                           none    0.000000
22                       Pembroke    0.809197
23                Mexican_hairless   0.330741
24                      Chihuahua    0.793469
25                         kuvasz    0.309706
26                           none    0.000000
27                 French_bulldog    0.999201
28                            pug    0.943575
29                Blenheim_spaniel   0.972494
...                           ...         ...
1887                         none    0.000000
1888      black-and-tan_coonhound    0.529139
1889                    Chihuahua    0.978108
1890  soft-coated_wheaten_terrier    0.326467
1891                         none    0.000000
1892                     malamute    0.336874
1893             golden_retriever    0.858744
1894                  Newfoundland   0.278407
1895                         none    0.000000
1896                   Maltese_dog   0.857531
1897             Italian_greyhound   0.176053
1898                         none    0.000000
1899                         none    0.000000
```

```
1900              English_setter    0.298617
1901                       Lhasa    0.582330
1902                  bloodhound    0.195217
1903                         pug    0.489814
1904                 Walker_hound    0.260857
1905                Gordon_setter    0.503672
1906             golden_retriever    0.775930
1907             miniature_poodle    0.201493
1908             golden_retriever    0.007959
1909                         chow    0.692517
1910                         none    0.000000
1911          Bernese_mountain_dog    0.651137
1912            miniature_pinscher    0.560311
1913           Rhodesian_ridgeback    0.408143
1914               German_shepherd    0.596461
1915                       redbone    0.506826
1916       Welsh_springer_spaniel    0.465074

[1917 rows x 14 columns]
```

In [221]:
```
df.head()
```

Out[221]:
```
             tweet_id           timestamp             source  \
0  892420643555336193 2017-08-01 16:23:56  Twitter for iPhone
1  892177421306343426 2017-08-01 00:17:27  Twitter for iPhone
2  891815181378084864 2017-07-31 00:18:03  Twitter for iPhone
3  891689557279858688 2017-07-30 15:58:51  Twitter for iPhone
4  891327558926688256 2017-07-29 16:00:24  Twitter for iPhone

                                                text  \
0  This is Phineas. He's a mystical boy. Only eve...
1  This is Tilly. She's just checking pup on you...
2  This is Archie. He is a rare Norwegian Pouncin...
3  This is Darla. She commenced a snooze mid meal...
4  This is Franklin. He would like you to stop ca...

                                     expanded_urls  rating_numerator  \
0  https://twitter.com/dog_rates/status/892420643...              13.0
1  https://twitter.com/dog_rates/status/892177421...              13.0
2  https://twitter.com/dog_rates/status/891815181...              12.0
3  https://twitter.com/dog_rates/status/891689557...              13.0
4  https://twitter.com/dog_rates/status/891327558...              12.0

   rating_denominator      name dog_stages  retweet_count  favorite_count  \
0                  10   Phineas        NaN           8853           39467
1                  10     Tilly        NaN           6514           33819
2                  10    Archie        NaN           4328           25461
3                  10     Darla        NaN           8964           42908
```

```
4                  10  Franklin        NaN           9774           41048

                                          jpg_url                breed  \
0  https://pbs.twimg.com/media/DGKD1-bXoAAIAUK.jpg                 none
1  https://pbs.twimg.com/media/DGGmoV4XsAAUL6n.jpg            Chihuahua
2  https://pbs.twimg.com/media/DGBdLU1WsAANxJ9.jpg            Chihuahua
3  https://pbs.twimg.com/media/DF_q7IAWsAEuuN8.jpg  Labrador_retriever
4  https://pbs.twimg.com/media/DF6hr6BUMAAzZgT.jpg               basset

   confidence
0    0.000000
1    0.323581
2    0.716012
3    0.168086
4    0.555712
```

In [222]:
```
# making copy of master dataframe
df1 = df.copy()
```

In [223]:
```
# checking for null values
df1.isnull().sum()
```

Out[223]:
```
tweet_id                 0
timestamp                0
source                   0
text                     0
expanded_urls            0
rating_numerator         0
rating_denominator       0
name                     0
dog_stages            1614
retweet_count            0
favorite_count           0
jpg_url                  0
breed                    0
confidence               0
dtype: int64
```

In [224]:
```
# checking duplicated values
df1[df1.duplicated()]
```

Out[224]:
```
Empty DataFrame
Columns: [tweet_id, timestamp, source, text, expanded_urls, rating_numerator, rating_d
Index: []
```

In [225]:
```
#dropping dog_stage column
df1.drop(['dog_stages'], axis = 1, inplace=True)
```

In [226]:
```
df1.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 1917 entries, 0 to 1916
Data columns (total 13 columns):
tweet_id              1917 non-null int64
timestamp             1917 non-null datetime64[ns]
source                1917 non-null object
text                  1917 non-null object
expanded_urls         1917 non-null object
rating_numerator      1917 non-null float64
rating_denominator    1917 non-null int64
name                  1917 non-null object
retweet_count         1917 non-null int64
favorite_count        1917 non-null int64
jpg_url               1917 non-null object
breed                 1917 non-null object
confidence            1917 non-null float64
dtypes: datetime64[ns](1), float64(2), int64(4), object(6)
memory usage: 209.7+ KB
```

In [227]:
```
df1.describe()
```

Out[227]:
```
           tweet_id  rating_numerator  rating_denominator  retweet_count  \
count  1.917000e+03       1917.000000         1917.000000    1917.000000
mean   7.352520e+17         12.224716           10.490871    2721.184664
std    6.733677e+16         42.187551            6.948704    4656.052337
min    6.660209e+17          0.000000            2.000000      16.000000
25%    6.755224e+17         10.000000           10.000000     618.000000
50%    7.079958e+17         11.000000           10.000000    1336.000000
75%    7.869631e+17         12.000000           10.000000    3154.000000
max    8.924206e+17       1776.000000          170.000000   79515.000000

       favorite_count   confidence
count     1917.000000  1917.000000
mean      8736.636411     0.464178
std      12095.233439     0.340148
min         81.000000     0.000000
25%       1889.000000     0.139613
50%       4026.000000     0.457117
75%      11203.000000     0.776612
max     132810.000000     0.999956
```

1.9.28 checking for outliers

1.9.29 code

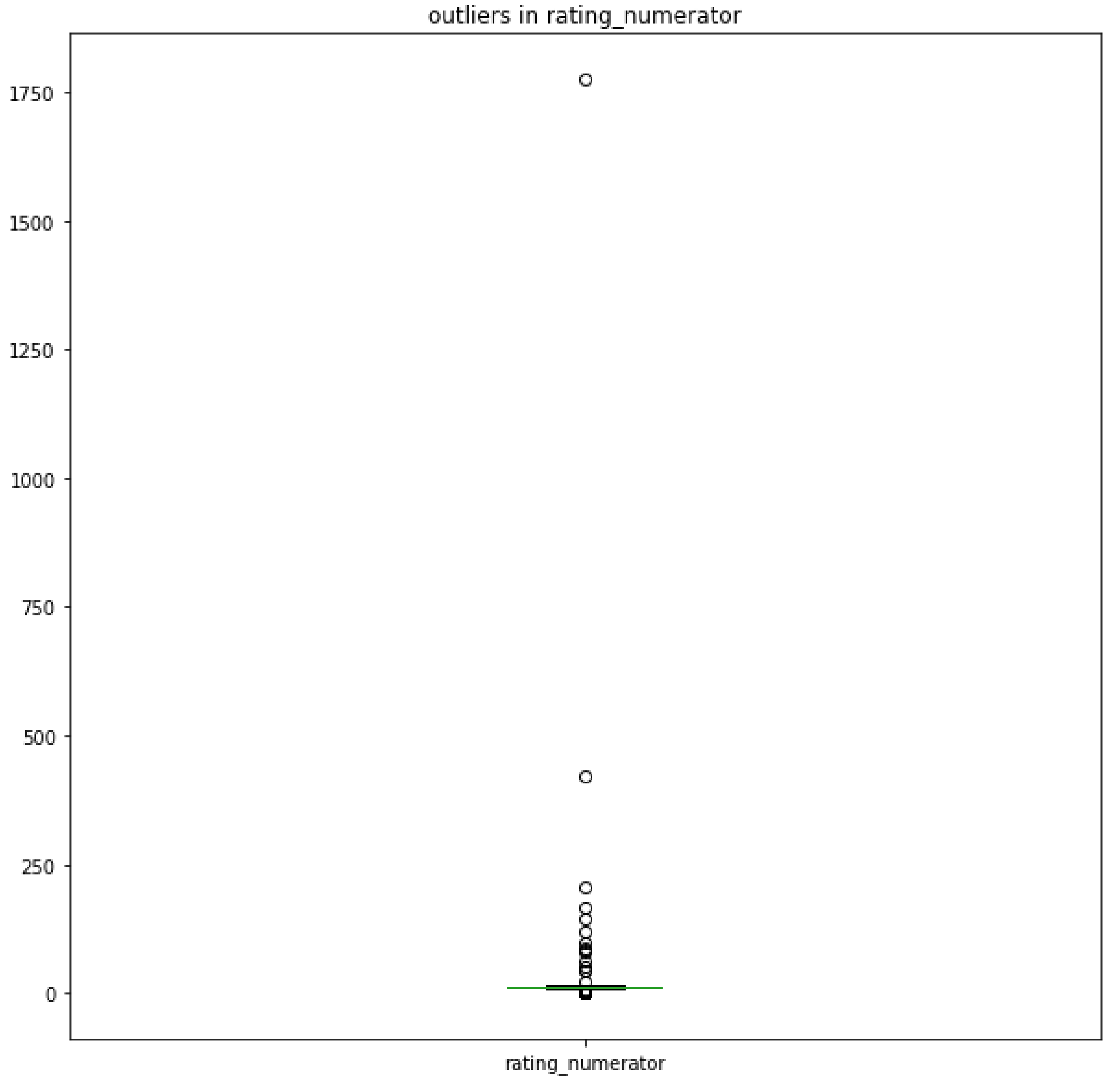
In [228]:
```
# checking for outliers
def get_outlier_index(df, col):
    q1 = df[col].quantile(0.25)
```

```
    q3 = df[col].quantile(0.75)
    fs = q3-q1
    limit = 1.5*fs
    lower_l = q1 - limit
    upper_l = q3 + limit
    ls = df.index[(df[col] < lower_l) | (df[col] > upper_l)]
    return ls
```

In [229]:
```
# checking for outliers
df1[(df1['rating_denominator'] < 10) | (df1['rating_denominator'] > 10)].index.value_c
```

Out[229]: 17

In [230]:
```
df1['rating_numerator'].plot(kind='box', figsize=(10,10));
plt.title('outliers in rating_numerator');
```



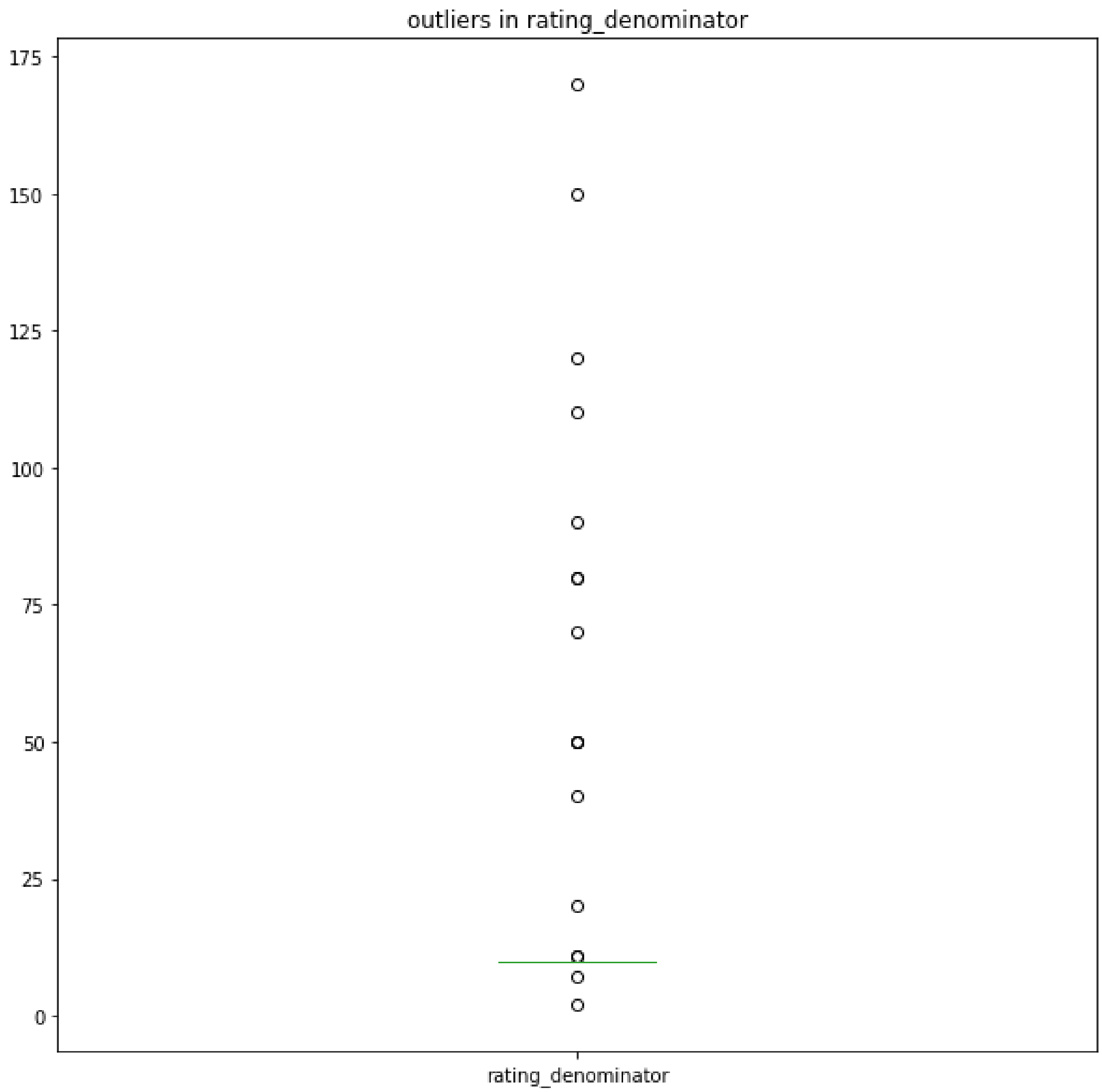
In [231]:
```
# getting index of outliers
# outlier_index = df[(df['rating_denominator'] < 10) | (df['rating_denominator'] > 10)
```

In [232]:
```
denominator_outlier_index = get_outlier_index(df1, 'rating_denominator')
```

In [233]:
```
drop(df1, denominator_outlier_index, 0)
```

In [234]:
```
#df['rating_denominator']
```

In [235]:
```
df['rating_denominator'].plot(kind='box', figsize=(10,10));
plt.title('outliers in rating_denominator');
```



1.9.30 Test

In [236]:
```
# checking for outliers
df1[(df1['rating_denominator'] < 10) | (df1['rating_denominator'] > 10)].index.value_c
```

```
Out[236]: 0
```

1.9.31 checking for outliers in rating_numerator

1.9.32 code

```
In [237]: df1[(df1['rating_numerator'] > 17)].index.value_counts().sum()

Out[237]: 2

In [238]: numerator_outlier_index = get_outlier_index(df1, 'rating_numerator')
          numerator_outlier_index

Out[238]: Int64Index([ 164,  219,  507,  529,  620,  640,  675,  697,  710,  719,
                       ...
                       1877, 1883, 1887, 1891, 1895, 1899, 1903, 1910, 1912, 1913],
                      dtype='int64', length=113)

In [239]: df1['rating_numerator'].loc[219]

Out[239]: 0.0

In [240]: drop(df1,numerator_outlier_index,axis=0 )
```

1.9.33 test

```
In [241]: df1.describe()

Out[241]:            tweet_id  rating_numerator  rating_denominator  retweet_count  \
          count  1.787000e+03       1787.000000              1787.0    1787.000000
          mean   7.384549e+17         10.924331                10.0    2818.611080
          std    6.796859e+16          1.556103                 0.0    4784.143225
          min    6.660209e+17          7.000000                10.0      16.000000
          25%    6.767941e+17         10.000000                10.0     620.500000
          50%    7.116948e+17         11.000000                10.0    1402.000000
          75%    7.928288e+17         12.000000                10.0    3275.500000
          max    8.924206e+17         14.000000                10.0   79515.000000

                 favorite_count   confidence
          count     1787.000000  1787.000000
          mean      9121.197538     0.482491
          std      12387.160468     0.334676
          min         81.000000     0.000000
          25%       2053.500000     0.181855
          50%       4229.000000     0.482452
          75%      11654.500000     0.788406
          max     132810.000000     0.999956
```

1.10 Analyzing and Visualizing Data

In this section, analyze and visualize your wrangled data. You must produce at least **three (3) insights and one (1) visualization.**

1.10.1 Question 1: which mobile device is the mostly used by usersers of

```
In [242]: # getting value count for each tweet sources which also indicates device used.
          user_sources = df1['source'].value_counts()
          user_sources

Out[242]: Twitter for iPhone    1751
          Twitter Web Client      27
          TweetDeck                9
          Name: source, dtype: int64

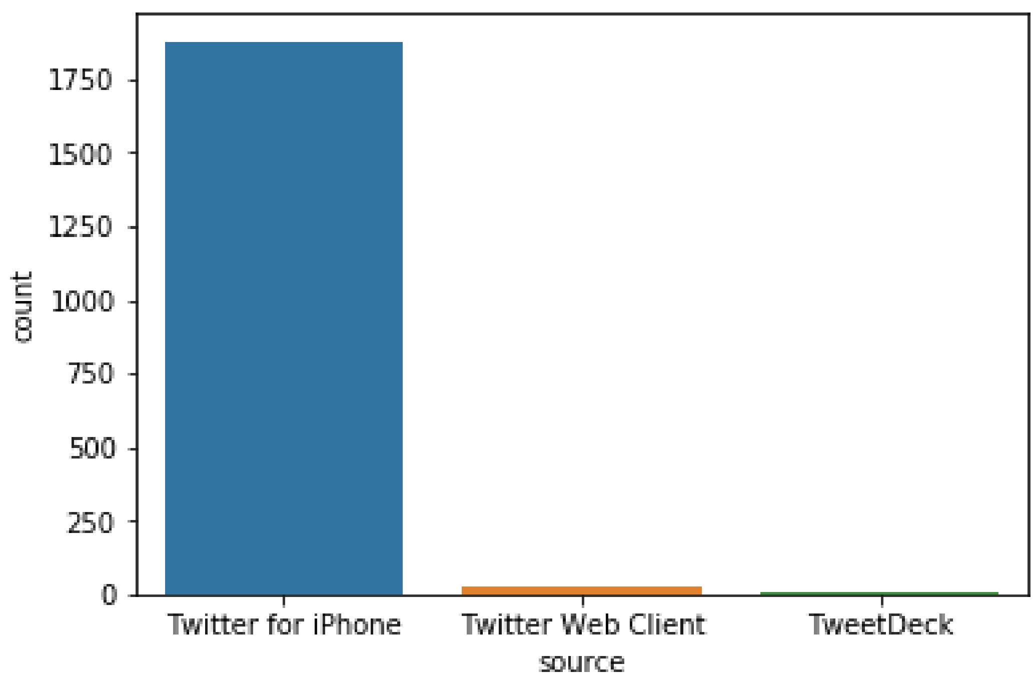
In [243]: # coverting the values in user_source into an array to know by how much is one source
          s = np.array(user_sources)
          s

Out[243]: array([1751,   27,    9])

In [244]: # estimating by how much iphone source is more than others
          iphone_by_how_much= s[0]/(s[1]+s[2])
          iphone_by_how_much

Out[244]: 48.638888888888886

In [245]: # constructing barchart for most used source
          sb.countplot(data= df1, x = df['source']);
```



iphone is the most used device by over fourty time more than other devices

1.10.2 Question 2: which is month has the most engagement over the three years period

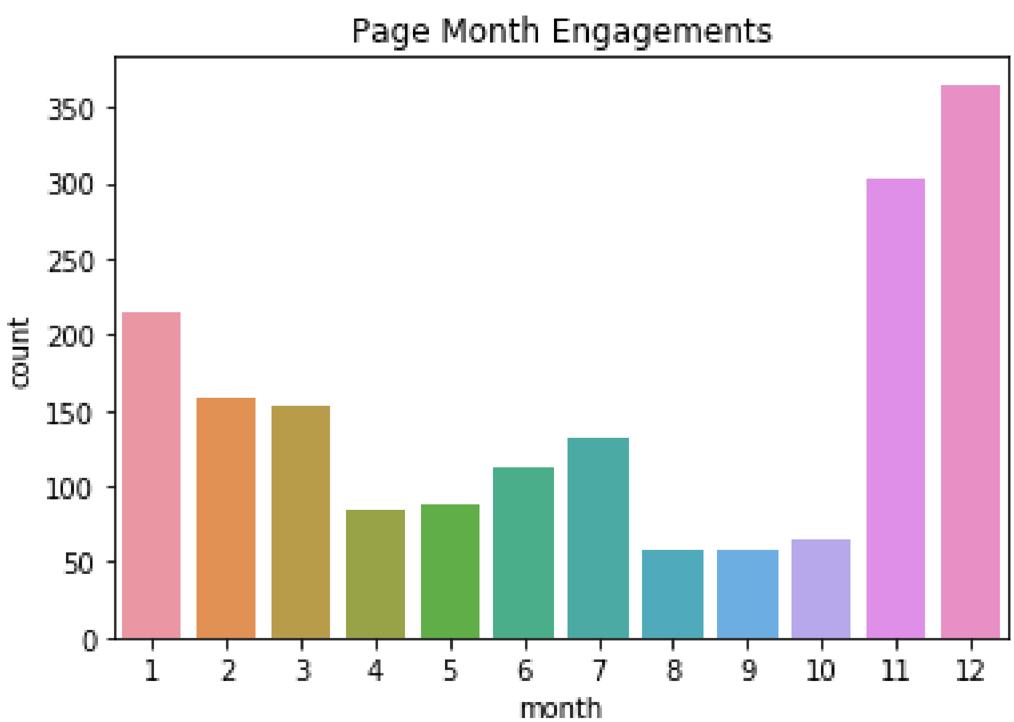
In [ ]:

In [246]:
```
# extracting months and creating month column
df1['month'] = df1['timestamp'].dt.month
```

In [247]:
```
count = df1['month'].value_counts()
count
```

Out[247]:
```
12    365
11    302
1     214
2     158
3     153
7     132
6     112
5      87
4      84
10     65
9      58
8      57
Name: month, dtype: int64
```

In [248]:
```
# constructing barchart for page engagement
sb.countplot(data= df1, x =df1['month']);
plt.title('Page Month Engagements');
```



most engagement happens towards the end of the year with December having highest engagement.

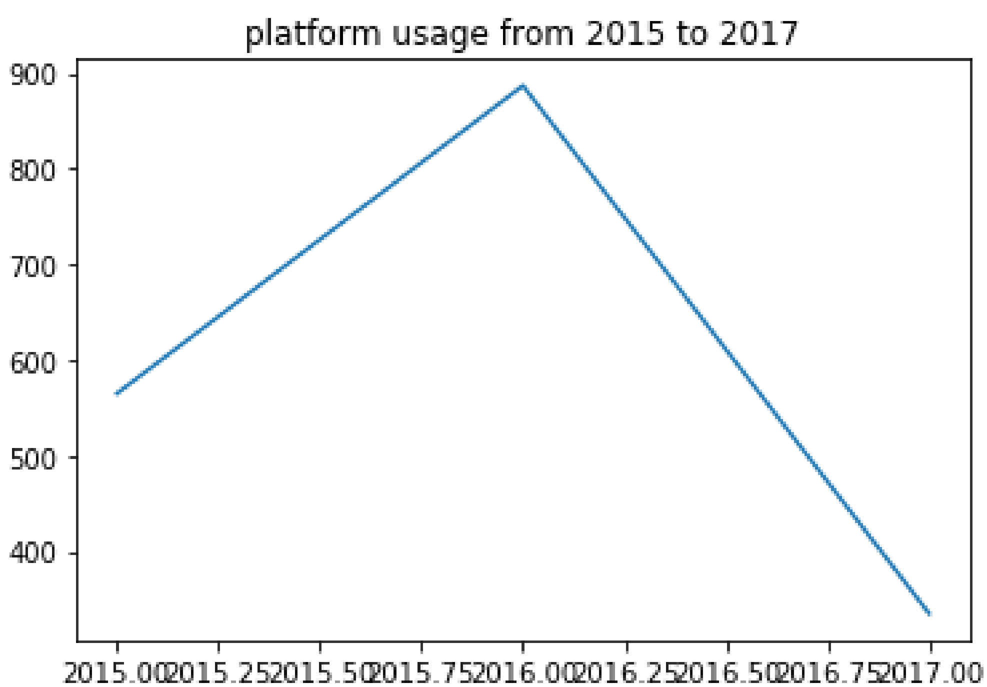
1.10.3 Question 3: what is the page usage (trend) over the three years

In [249]:
```
# extracting year and creating year column
df1['year'] = df1['timestamp'].dt.year
```

In [250]:
```
df1['year'].value_counts(sort=False)
```

Out[250]:
```
2015    565
2016    887
2017    335
Name: year, dtype: int64
```

In [251]:
```
# plottingline graph for platform usgae acroos three years peroid
plt.plot(df1['year'].value_counts(sort=False));
plt.title('platform usage from 2015 to 2017');
```



page engagement declined from 2016 to 2017

1.10.4 Insights:

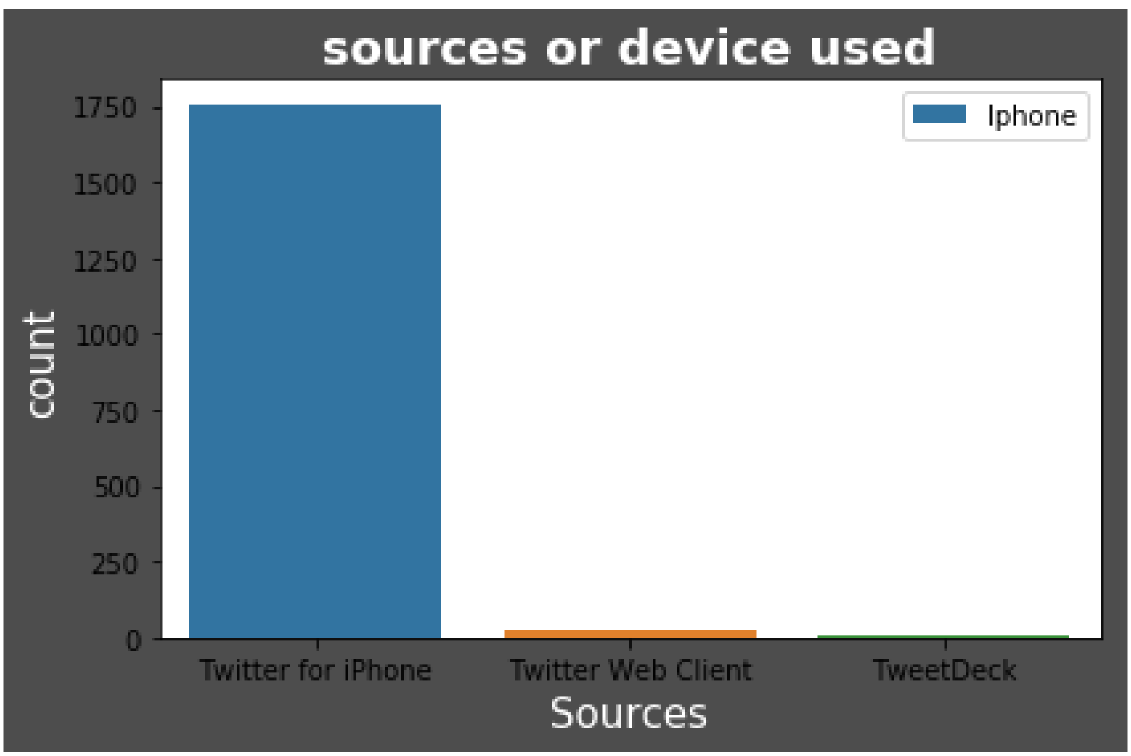
1. majority of the users are iphone users. Iphone users appears to be over forty times more than other sources or deviced used.

2. december had most engagement across the years

3. usage of the page has been declining

1.10.5 Visualization

```
In [252]: labels = ['Iphone','Web', 'Deck']
          ax = sb.countplot(x='source', dodge=False, data=df1)
          ax.set_title('sources or device used', fontsize=18, fontweight='bold', color='white')
          ax.set_xlabel('Sources', fontsize=15, color='white')
          ax.set_ylabel('count', fontsize=15, color='white')
          ax.figure.set_facecolor('0.3')
          plt.legend(labels=labels)
          plt.tight_layout()
          plt.show()
```



1.10.6 Refrences

- stackoverflow
- youtube

```
In [ ]:
```

```
In [ ]:
```